

uCloudlink Interface Specification Document for Business Layer (Business Customer)

Version 2.9
23 July 2020

Shenzhen uCloudlink Technology Co., Ltd

Shenzhen uCloudlink reserves all rights to the revision and final interpretation of the specifications.

1 Introduction and Related Docs.....	9
1.1 Background.....	9
1.2 Glossary.....	9
1.3 Related Documents.....	9
2 Main Description.....	10
2.1 Involved System.....	10
2.2 Functional Architecture.....	10
2.3 Interface Description.....	11
2.4 Business Scenarios.....	11
2.4.1 Business customer login in and login out.....	11
2.4.2 Manage sub-users and it's device.....	12
2.4.3 Manage sub-users' tariff.....	12
2.4.4 Manage sub-users' balance.....	13
2.4.5 Query sub-users' information.....	14
2.5 Access Control.....	14
3 Interface for Business Customer.....	15
3.1 List of Interface.....	15
3.2 Technical Proposal.....	15
3.2.1 Interface Type.....	16
3.2.2 Message Type.....	16
3.2.3 Business Customer Account.....	16
3.2.4 MVNO Code.....	16
3.2.5 Login and Request.....	16
3.2.6 Partner Service Domain limitation.....	16
3.2.7 Result code.....	17
3.2.8 Call Entry 26.....	26
3.2.9 Content-Type:application/json.....	26
3.2.10 Initial Parameter Provided to Partner.....	26
3.2.11 Access_token.....	26
3.2.12 Test Devices.....	27
3.2.13 Code Demo for invoke interface (JAVA).....	27
3.2.14 Code Demo for invoke interface (PHP).....	29
4 Order Mode's Primary Part (Default provide).....	30
4.1 Business customer login in.....	30
4.1.1 Overview 30.....	30
4.1.2 Main Description.....	30
4.1.3 Input Parameter.....	30
4.1.4 Output Parameter.....	30
4.2 Business customer login out.....	31
4.2.1 Overview 31.....	31
4.2.2 Main Description.....	31
4.2.3 Input Parameter.....	31
4.2.4 Output Parameter.....	32
4.3 Batch create sub-user of business customer.....	32
4.3.1 Overview 32.....	32
4.3.2 Main Description.....	32
4.3.3 Input Parameter.....	32
4.3.4 ChildUserVO Entity Description.....	33
4.3.5 Output Parameter.....	33
4.3.6 retData Entity Description.....	34
4.3.7 ErrorVO Entity Description.....	34
4.4 Query sub-user list of business customer.....	35
4.4.1 Overview 35.....	35
4.4.2 Main Description.....	35

4.4.3 Input Parameter.....	35
4.4.4 Output Parameter.....	35
4.4.5 PageData Entity Description.....	36
4.4.6 SubUserVO Entity Description.....	36
4.5 Query available offer list for sale.....	37
4.5.1 Overview 37	
4.5.2 Main Description.....	37
4.5.3 Input Parameter.....	37
4.5.4 Output Parameter.....	38
4.5.5 PageData Entity Description.....	38
4.5.6 GoodsVO Entity Description.....	39
4.6 Create order for sub-user.....	40
4.6.1 Overview 40	
4.6.2 Main Description.....	40
4.6.3 Input Parameter.....	40
4.6.4 OrderItemVO Entity Description.....	41
4.6.5 Output Parameter.....	41
4.7 Query offer list of sub-user.....	42
4.7.1 Overview 42	
4.7.2 Main Description.....	42
4.7.3 Input Parameter.....	42
4.7.4 Output Parameter.....	43
4.7.5 PageData Entity Description.....	44
4.7.6 OrderRelationVO Entity Description.....	44
5 Order Mode's Secondary Part (Open only after application).....	47
5.1 Stop and start sub-user (batch).....	47
5.1.1 Overview 47	
5.1.2 Main Description.....	47
5.1.3 Input Parameter.....	47
5.1.4 Output Parameter.....	47
5.1.5 retData Entity Description.....	48
5.1.6 ErrorVO Entity Description.....	48
5.2 Delete and resume sub-user (batch).....	48
5.2.1 Overview 48	
5.2.2 Main Description.....	49
5.2.3 Input Parameter.....	49
5.2.4 Output Parameter.....	49
5.2.5 retData Entity Description.....	50
5.2.6 ErrorVO Entity Description.....	50
5.3 Reset sub-user password.....	50
5.3.1 Overview 50	
5.3.2 Main Description.....	50
5.3.3 Input Parameter.....	51
5.3.4 Output Parameter.....	51
5.4 Bind device with sub-user.....	51
5.4.1 Overview 51	
5.4.2 Main Description.....	52
5.4.3 Input Parameter.....	52
5.4.4 Output Parameter.....	52
5.5 Unbind device with sub-user (batch).....	53
5.5.1 Overview 53	
5.5.2 Main Description.....	53
5.5.3 Input Parameter.....	53
5.5.4 Output Parameter.....	53
5.5.5 retData Entity Description.....	54
5.5.6 ErrorVO Entity Description.....	54
5.6 Query device binding Information.....	54
5.6.1 Overview 54	

5.6.2 Main Description.....	54
5.6.3 Input Parameter.....	55
5.6.4 Output Parameter.....	55
5.6.5 TerminalActivationVo Entity Description.....	56
5.7 Lock and unlock device (batch).....	56
5.7.1 Overview 56	
5.7.2 Main Description.....	56
5.7.3 Input Parameter.....	57
5.7.4 Output Parameter.....	57
5.7.5 retData Entity Description.....	57
5.7.6 ErrorVO Entity Description.....	58
5.8 Top up for sub-user.....	58
5.8.1 Overview 58	
5.8.2 Main Description.....	58
5.8.3 Input Parameter.....	58
5.8.4 Output Parameter.....	59
5.8.5 BalanceVO Entity Description.....	59
5.9 Query top up list of sub-user.....	59
5.9.1 Overview 59	
5.9.2 Main Description.....	60
5.9.3 Input Parameter.....	60
5.9.4 Output Parameter.....	60
5.9.5 PageData Entity Description.....	61
5.9.6 BalanceChargeVO Entity Description.....	61
5.10 Query account detail of sub-user.....	61
5.10.1 Overview.....	61
5.10.2 Main Description.....	62
5.10.3 Input Parameter.....	62
5.10.4 Output Parameter.....	62
5.10.5 PageData Entity Description.....	63
5.10.6 BalanceDetailVO Entity Description.....	63
5.11 Batch query sub-user list of business customer.....	64
5.11.1 Overview.....	64
5.11.2 Main Description.....	64
5.11.3 Input Parameter.....	64
5.11.4 Output Parameter.....	65
5.11.5 QuerySubUserListByBatchVO Entity Description.....	65
5.12 Operate offer for sub-user.....	66
5.12.1 Overview.....	66
5.12.2 Main Description.....	66
5.12.3 Input Parameter.....	66
5.12.4 GoodsVo Entity Description.....	67
5.12.5 Output Parameter.....	67
5.13 Query offer list of sub-user by orderId.....	68
5.13.1 Overview.....	68
5.13.2 Main Description.....	68
5.13.3 Input Parameter.....	68
5.13.4 Output Parameter.....	69
5.13.5 OrderRelationSubVO Entity Description.....	69
5.14 Unsubscribe order for sub-user.....	69
5.14.1 Overview.....	69
5.14.2 Main Description.....	70
5.14.3 Input Parameter.....	70
5.14.4 Output Parameter.....	70
5.15 Operate OrderRelation for sub-user.....	71
5.15.1 Overview.....	71
5.15.2 Main Description.....	71
5.15.3 Input Parameter.....	71
5.15.4 GoodsVo Entity Description.....	71

5.15.5 Output Parameter.....	72
5.16 Disable OrderRelation for sub-user.....	72
5.16.1 Overview.....	72
5.16.2 Main Description.....	72
5.16.3 Input Parameter.....	72
5.16.4 GoodsVo Entity Description.....	73
5.16.5 Output Parameter.....	73
5.17 Update OrderRelation for sub-user.....	74
5.17.1 Overview.....	74
5.17.2 Main Description.....	74
5.17.3 Input Parameter.....	74
5.17.4 GoodsVo Entity Description.....	75
5.17.5 Output Parameter.....	75
6 Process for APIs Application.....	76
7 Test Case for Reference.....	77
7.1 Test Step.....	77
7.2 Primary Part's Test Scenarios.....	78
7.2.1 Business customer login in.....	78
7.2.2 Business customer login out.....	79
7.2.3 Batch create sub-user of business customer.....	80
7.2.4 Query sub-user list of business customer.....	81
7.2.5 Query available offer list for sale.....	83
7.2.6 Create order for sub-user.....	85
7.2.7 Query offer list of sub-user.....	86
7.3 Secondary Part's Test Scenarios.....	88
7.3.1 Stop and start sub-user (batch).....	88
7.3.2 Delete and resume sub-user (batch).....	89
7.3.3 Reset sub-user password.....	90
7.3.4 Bind device with sub-user.....	91
7.3.5 Unbind device with sub-user (batch).....	92
7.3.6 Query device binding Information.....	93
7.3.7 Lock and unlock device (batch).....	94
7.3.8 Top up for sub-user.....	95
7.3.9 Query top up list of sub-user.....	96
7.3.10 Query account detail of sub-user.....	98
7.3.11 Batch query sub-user list of business customer.....	100
7.3.12 Operate offer for sub-user.....	101
7.3.13 Query offer list of sub-user by orderId.....	102
7.3.14 Unsubscribe order for sub-user.....	102
7.3.15 Operate OrderRelation for sub-user.....	103
7.3.16 Disable OrderRelation for sub-user.....	104
7.3.17 Update OrderRelation for sub-user.....	105

Revision record			
Version No	Changes	By	Date
1.0	Created new version	liwei	27-May-16
1.1	Added new interface and business scenarios description	liwei	17-Aug-16
1.2	Add mvnoCode for 'business customer login in' interface in section 3.3.3	liwei	22-Sep-16
1.3	Fix a mistake in section 3.7.2 about Call Entry;	liwei	9-Oct-16
	Fix a mistake in section 3.11.2 Call Entry;		
	Replace http://open.ucloudlink.com/XXX into https://saas.ucloudlink.com/bss		
	Add new section:		
	3.2.8 Call Entry		
	3.2.9 Content-Type:application/json		
	3.2.10 Initial Parameter Provided to Partner		
	3.2.11 Access_token		
	3.2.12 Test Devices		
	3.2.13 Code Demo for invoke interface (JAVA)		
	3.2.14 Code Demo for invoke interface (PHP)		
	- Add new section 5 Test Case - Delete parameter example in section 3 Interface for Business Customer		
1.4	3.14.3 Add new Input parameter goodsType,categoryCode change value range	shuciliang	21-Feb-2017
	3.6.4.2 ChildUserVO Add Input parameter registerType		
	3.14.6.2 GoodsVO Add Output parameter attrMap		
	periodUnit		
	3.18.3 Add new Input parameter goodsType		
	3.18.6.2 OrderRelationVO Add Output parameter goodsType		
	periodUnit		
	attrMap		
	3.20.3 Body of File Update		
	Login_time Remark		
1.5	Logout_time Remark	shucailiang	23-Feb-17
	TradType Remark		
1.6	3.18.3 Body of File Update	lixuegang	15-Mar-17
	Fix a mistake about goodsType		

	● Add new Section		
	2.5 Access Control of interface		
	3.14.6.3 attrMap Attribute		
	3.18.6.3 attrMap Attribute		
	3.21 Batch query sub-user list of business customer		
	3.22 Operate offer for sub-user		
	● Add new section 3 Test Case		
1.7	● 3.20.3 Body of File Update Example	shucailiang	23-Mar-17
	● 3.22.3 operate Type add value		
	● 3.18.6.2 Entity Attribute add accumulatedFlowList		
1.8	● 3.22.3 Input Parameter update	shucailiang	28-Mar-2017
	● 3.22.4 GoodsVo Entity description Add		
	● 5.2.20.2 Input Parameter update		
1.9	● Modify some known errors	shucailiang	22-May-2017
2.0	● Modify some known errors ● Modify section 3.2.7 to delete repeated ResultCode 01020063 ● Modify section 3.15.5 to Add orderId in data	shucailiang	9-Aug-2017
2.3	● Add new section 3.23 ● Modify section 3.18.3 ADD orderId	shucailiang	1- Aug-2018
	● Add valid period for access token	wumengjiao	23- Aug- 2018
2.4	● Modify some description ● Change the order of some interfaces ● Add new interfaces 5.15/6.6 Operate self-define offer for sub-user 5.16/6.8 Invalid offer for sub-user - Missing examples of these two interfaces	lihongwei	16- Oct-2018
2.4.1	Modify the name and ID of interface 4.6/4.8/5.15/6.6	lihongwei	26- Oct-2018
2.4.2	Add examples of interface 4.6/4.8/5.15/5.16	lihongwei	17-Jan-2019
	Delete the interface for the synchronous purchase package		
2.5	4.3.4 Add orgCode	shucailiang	20-Feb-2018
	Add new section 5.17		
2.5.1	Modify the operate type description of top up interface	lihongwei	14- Mar-2019
2.6	Add DIY package, customers can choose the coverage country in the package	shucailiang	14- July-2019
2.7	Add a new interface 5.18, transform device to a target org	shucailiang	23- Aug-2019

2.8	Add a new interface 5.19, Query online user count and card count	shucailiang	19-Nov-2019
2.9	Delete interface “transform device to a target org” and “Query online user count and card count”	lihongwei	23- July-2020

Confidential & uCloudlink

1 Introduction and Related Docs

1.1 Background

The document is used for third party partner to invoke uCloudlink interfaces for business layer. The document includes service flow, interface definition and use guide. This document describes in detail how uCloudlink Application Programming Interface (API) platform provides basic services and how to define and use a list of interfaces. If data invoked by an interface involves users' private information, please comply with our Confidentiality Agreement and do not reveal or disclose the information to the public.

Shenzhen uCloudlink reserves all rights to the revision and final interpretation of the specifications.

1.2 Glossary

NA

1.3 Related Documents

NA

2 Main Description

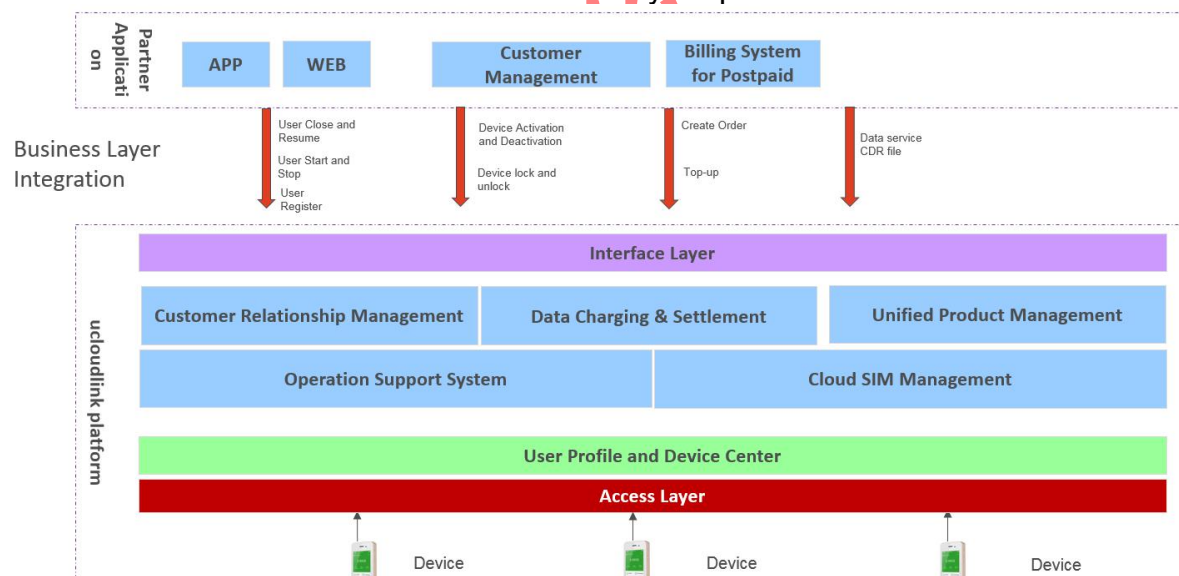
2.1 Involved System

These are three elements involved in the whole service flow for open interface in access layer:

- **Partner System:** Partner system includes back-end system for example customer management, billing management for postpaid; Partner system also includes APP/WEB which provide end-user self-services.
- **Interface layer of uCloudlink Platform:** The interface layer is provided by uCloudlink platform which receive request from partner system and forward to service layer of uCloudlink platform. Partner need to have administrator account and password of each business customer, and uCloudlink has open interface roles for these account.
- **Business Layer of uCloudlink Platform:** The business layer is provided by uCloudlink platform which could process partner request and deal with the business. The business layer includes customer relationship management, data charging & settlement, unified product management. The balance for prepaid and data tariff is managed in uCloudlink Platform.

2.2 Functional Architecture

Below is functional architecture of business layer open for business customer:



In this case, partner will build its own app and web for consumer and business, or partner could integrate data service business into partner's existed app and web. Partner could provide personalized experience to customer.

Partner could build its own customer management portal to manage sub-users of each business customer, sub-users could be postpaid or prepaid. Partner's consumer customer could be deal as sub-users of a common business customer.

2.3 Interface Description

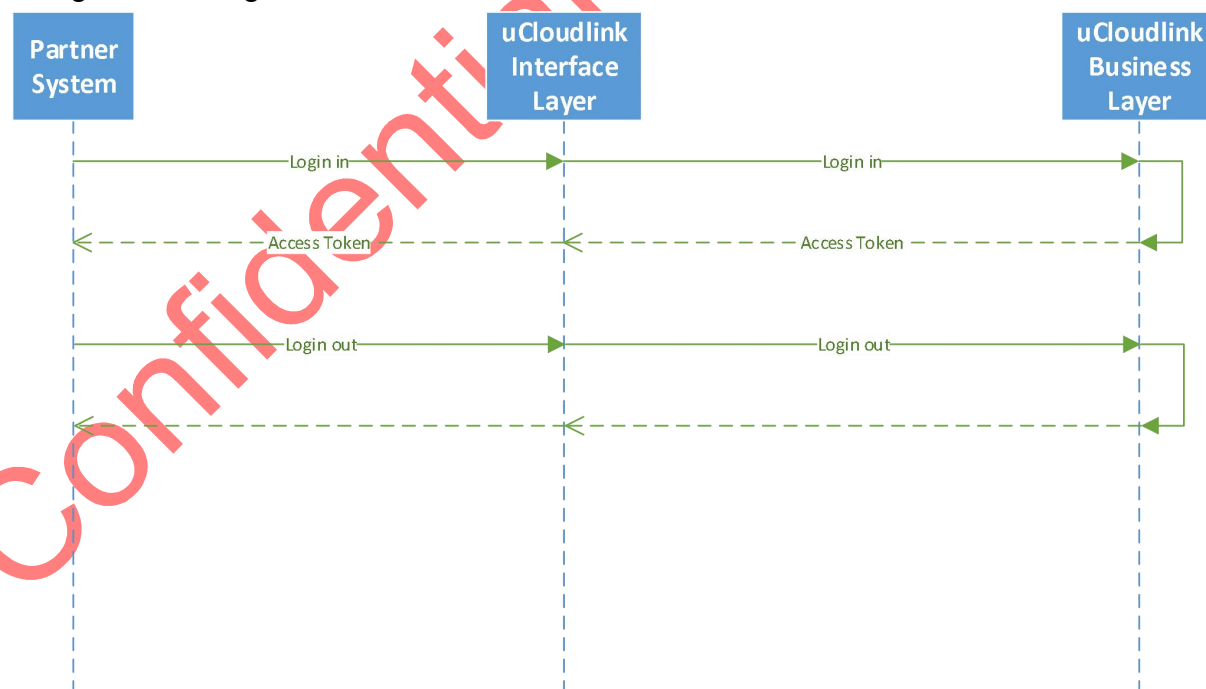
Interfaces opened in business layer are grouped in six kinds of APIs:

- APIs for business customer login and logout: including business customer login in, business customer login out
- APIs for manage sub-users and relation between device and sub-users: including query sub-user list of business customer, batch create sub-user of business customer, stop and start sub-user (batch), delete and resume sub-user (batch), reset sub-user password, query device binding Information, bind device with sub-user, unbind device with sub-user (batch), lock and unlock device (batch)
- APIs for manage sub-users' tariff: including query available offer list for sale, create order for sub-user
- APIs for manage sub-users' balance: including top up for sub-user, query top up list of sub-user
- APIs for query sub-users' information: including query offer list of sub-user, query account detail of sub-user

2.4 Business Scenarios

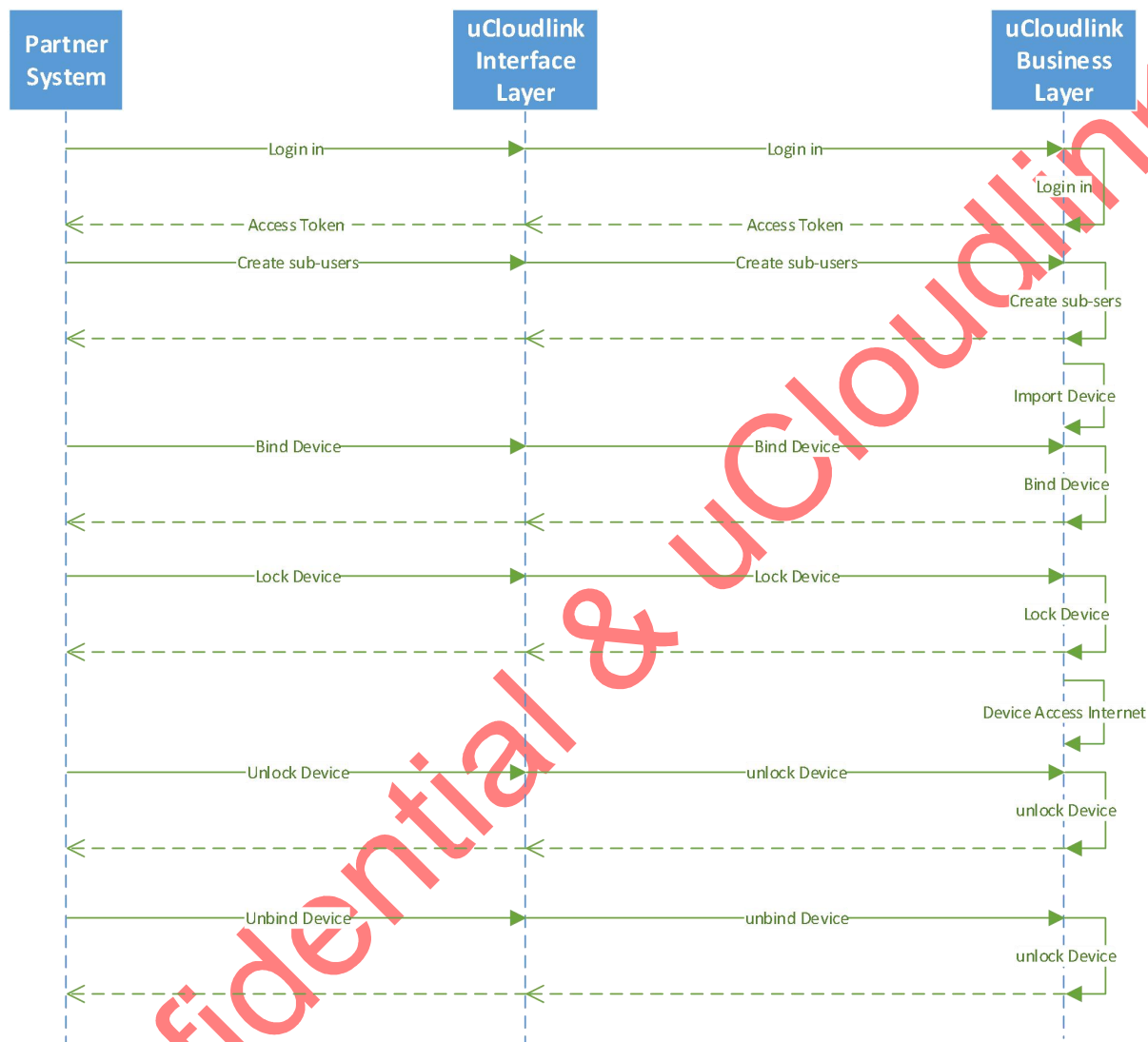
2.4.1 Business customer login in and login out

Before partner system call APIs to do register, device binding, top-up etc. partner system need login in as business customer first by JA-SIG Central Authentication Service following OAuth2.0 standard. After business customer login, partner system could manage sub-users with belong this business customer. If partner have different business customer, partner system should use different business customer account to login in and login out.



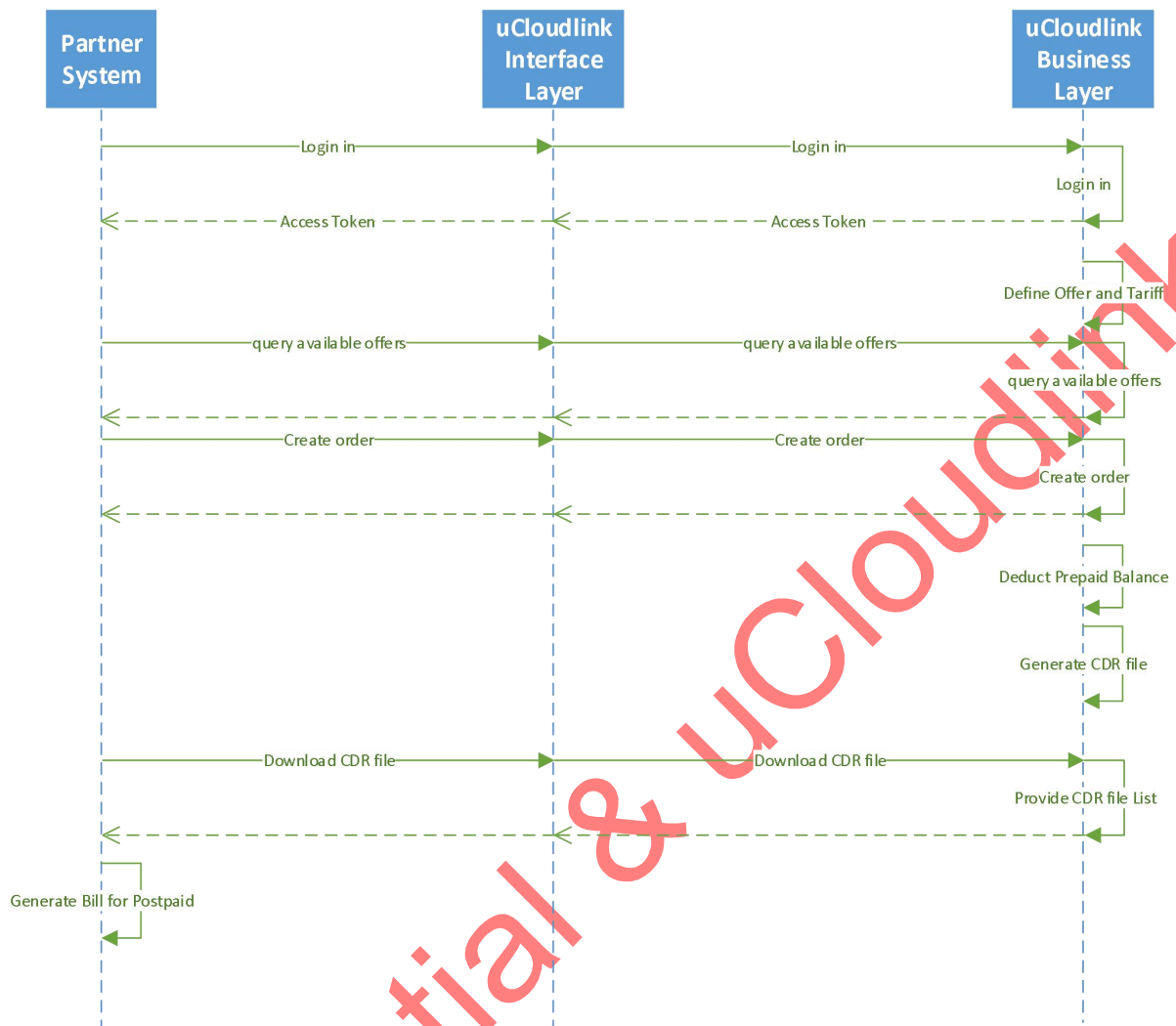
2.4.2 Manage sub-users and it's device

Before the device could access internet, it should have a sub-user account has already bind with the device. The relation between device and sub-users could be locked and unlocked. Partner system could create sub-users for a business customer and query sub-users list.



2.4.3 Manage sub-users' tariff

Before device access internet, the sub-account bind with device should be assign tariff first. The tariff need be defined as offer first in uCloudlink system by administrator using portal. Partner system could query available offers for the business customer, and create order to assign offer to sub-users. For prepaid customer, the tariff will reduce customer balance in uCloudlink system; for postpaid customer, the tariff could be defined as ZERO and set a usage limitation to control credit in uCloudlink system, then partner billing system could use CDR file to generate monthly bill.

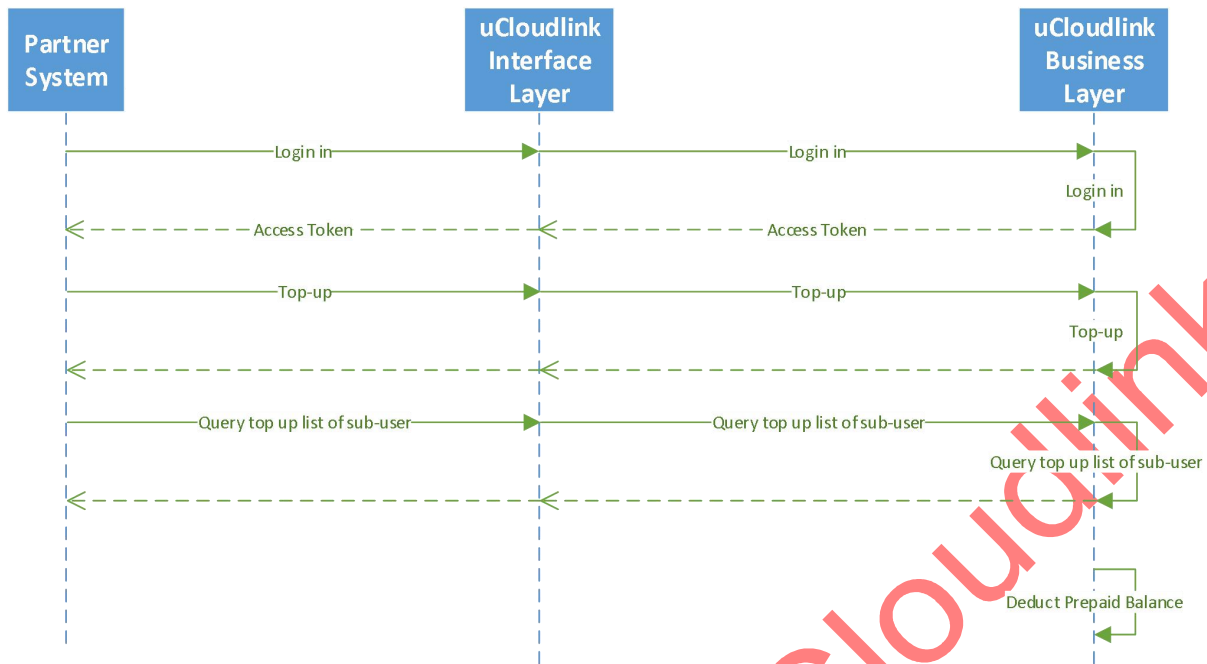


The offer and tariff which uCloudlink supported is list below:

- PAYG tariff which deduct prepaid balance.
- Data bundle with specified volume without speed limitation in a period (Customer need purchase data bundle and deduct balance. For postpaid customer, the price of data bundle could be defined as ZERO.
- Daily/Monthly Usage Bundle with specified volume, and do speed limitation when reach the specified volume in a period. For postpaid customer, the price of daily/monthly bundle could be defined as ZERO.

2.4.4 Manage sub-users' balance

For prepaid customer, the balance is managed in uCloudlink system. When device access internet using PAYG or sub-users purchase data bundle, it should directly deduct uncloudlink sub-users' balance in real time. So partner system need top-up prepaid customer balance when necessary.



2.4.5 Query sub-users' information

Partner system could query sub-users' account detail, offer list.

2.5 Access Control

The Access Control is the way of managing the access authority such as Call Frequency, Filtering the IP. If found abnormal call behavior, we have the right to close the interfaces. uCloudlink reserves the charge right for the access.

3 Interface for Business Customer

3.1 List of Interface

This mode is a normal ordering route by purchasing a package through the CreateOrder interface.

Primary Part (Default provide)		
NO.	Interface name	Interface ID
1	Business customer login in	GrpUserLogin
2	Business customer login out	GrpUserLogout
3	<u>Batch create sub-user of business customer</u>	BatchCreateGroupChild
4	Query sub-user list of business customer	QuerySubUserListInfo
5	Query available offer list for sale	QueryGrpOfferList
6	Create order for sub-user	GrpCreateOrder
7	Query offer list of sub-user	QueryUserOfferList
Secondary Part (Open only after application)		
NO.	Interface name	Interface ID
1	Stop and start sub-user (batch)	StopSubUser
2	Delete and resume sub-user (batch)	DeleteSubUser
3	Reset sub-user password	ResetPwdForSubUser
4	Bind device with sub-user	BindDevice
5	Unbind device with sub-user (batch)	BatchUnBindDevice
6	Query device binding Information	QueryBindingRelationInfo
7	Lock and unlock device (batch)	LockDevice
8	Top up for sub-user	TopUpForSubUser
9	Query top up list of sub-user	QueryTopUpListInfo
10	Query account detail of sub-user	QueryAccountDetailListInfo
11	Batch query sub-user list of business customer	QuerySubUserListByBatch
12	Operate offer for sub-user (Old)	SyncOrderRelationFromUP
13	Query offer list of sub-user by orderId	QueryOrderRelationByOrderId
14	Unsubscribe order for sub-user	CancelOrder
15	Operate OrderRelation for sub-user (New)	OperateOrderRelation
16	Disable OrderRelation for sub-user (New)	DisableOrderRelation
17	Update OrderRelation for sub-user (New)	UpdateOrderRelation

3.2 Technical Proposal

Interface Type: HTTP+ POST

Message Type: JSON format

Authentication Method: Login by business customer user name and password via interface provided by JA-SIG Central Authentication Service following OAuth2.0 standard, the platform will check whether the business customer has the role invoking the interface.

3.2.1 Interface Type

The interface type is HTTP+ POST, below is interface address example:

<https://saas.ucloudlink.com/bss/grp/noauth/GrpUserLogin>

Partner system need sent HTTPS+POST request to these address.

uCloudlink does not provide HTTP interface because of safety requirement.

3.2.2 Message Type

The content of HTTP+POST request is in JSON format, below is JSON example:

```
{
  "streamNo" : "UKGRP20151113161019000001",
  "clientId" : "bb18c96ca5fd4c82bca5788f9a0a4311",
  "clientSecret" : "563b2c3404d0321a149a0ade25b54c82",
  "userCode" : "8618520846179",
  "langType": "zh-CN",
  "password": "561b7301dc199e1fec5744ba5431f8d7",
  "partnerCode": "UKGRP",
  "mvnoCode": "MVNO"
}
```

3.2.3 Business Customer Account

Partner need to apply business customer account and password from uCloudlink offline, if partner has different business customer, partner need apply several business customer accounts. Normally one MVNO just need one common business customer account for its all consumer customer and sub-users of business customer, partner could group different customer in its owner system.

If different business customer in one MVNO have different basic PAYG tariffs for prepaid, partner system should apply different business customer accounts.

3.2.4 MVNO Code

Partner need to apply MVNO code from uCloudlink offline. Multiple business customer accounts could belong to one MVNO, one partner could apply multiple MVNO code if necessary.

3.2.5 Login and Request

Most of interfaces for business customer need login first, partner service could login by business customer account and password via interface provided by JA-SIG Central Authentication Service following OAuth2.0 standard, the platform will check whether the business customer has role invoking specified interface.

3.2.6 Partner Service Domain limitation

Partner need submit interface application with service domain which will directly invoke uCloudlink interface for business layer.

If partner has changed service domain of the service, partner need submit change request to uCloudlink a week before change.

uCloudlink will keep service domain in white list and allow partner service make request to uCloudlink interface server.

3.2.7 Result code

Below is result codes dictionary:

ResultCode	ResultDesc
00000000	Success
00000001	streamNo is null
00000002	partnerCode is null
00000003	loginCustomerId is null
00000004	partnerCode does not exist
00000005	loginCustomerId does not exist
00000006	accessToken is null
00000007	accessToken is expire
00000008	accessToken does not belong to the loginCustomerId
00000009	clientId is null
00000010	clientSecretId is null
00000011	clientId does not exist
00000012	clientSecretId does not exist
00000013	customerId is null
00000014	data already exists
00000015	sessionId is null
01010001	MCC is null
01010002	Rate is not exist
01010003	ID is null
01010004	ISO2 is null
01010005	MCC does exist
01010006	ISO2 is not exist
01010007	Name does exist
01010008	Continent is null
01010009	ISO4 is null
01010010	Continent is not exist
01010011	Country does not exist
01010012	ISO3 does exist
01010013	Key is null
01010014	Type is null
01010015	Name does exist
01010016	Id is null
01010017	Chinese is null
01010018	English is null
01010019	Key does exist
01010020	ID is null
01010021	id is not exist
01010022	Code is null
01010023	name is null
01010024	Data is null
01010025	Code does exist

01010027	id is not exist
01010028	Type is null
01010029	ExchRate is null
01010030	Rate is null
01010031	Type is null
01010032	Id is null
01010033	id is not exist
01010034	MVNO has no auth to paramCode
01010035	ParamCodeList is null
01010036	id is not exist
01010037	ParamCodeList is mistake
01010038	MVNO has no auth to paramCode
01010039	ParamCode is not exist
01010040	langType must not be null
01010041	ParamCode is null
01010042	id is not exist
01010043	Id is not exist
01020001	customerId is null
01020002	customer has not login
01020003	login customer has no auth to operate the data
01020004	username is null
01020005	new password is null
01020006	old password is null
01020007	old password is not right
01020008	id is null
01020009	id is not exist
01020010	Deleted customer failed, the customer is MVNO corporate
01020011	Deleted user failed, the user has binding a imei
01020020	currencyType is null
01020021	mvnoParentId is null
01020022	mvnoName is null or length above 30
01020023	mvnoCode is null or length is not equal 5
01020024	username length can not above 30
01020025	firstname is null
01020026	lastname is null
01020027	countryCode is null
01020028	phone is null
01020029	phone format is illegal
01020030	The format of Email is error
01020031	The length of remark must less 255
01020032	identificationType is null or illegal
01020033	The login user is not exist a customer
01020034	currencyType is not exist
01020035	mvnoParentId is not exist
01020036	mvnoName has exist
01020037	mvnoCode has exist
01020038	mvnoParentId is not belong login user
01020039	UserId is null
01020040	username {0} has exist

01020041	Operator database exception
01020042	mvnoState is null or illegal
01020043	The customer is not exist or delete or disable
01020044	customerId is not MVNO operator
01020045	Insufficient permissions to operator
01020046	customerId is not org operator
01020047	the customer type must be TERMINALUSER
01020048	the customer type must be OPERUSER
01020049	username must be a valid email address
01020050	username is not equals email
01020051	username must be a valid phone number
01020052	username is not equals countryCode add phone
01020053	registerType is illegal
01020054	verifymode is null
01020055	email is null
01020056	verifymode is illegal
01020057	{0} and {1} at least has one value
01020058	user password length between 6 and 20
01020059	customerLevel is null
01020060	registerType is null
01020061	the customer's organization level must equal or below the operator's organization level
01020062	customer must be org operUser
01020063	customerId is not common operator
01020066	No transfer data
01020067	Data is greater than 100
01020068	No transfer data
01020069	Data is greater than 100
01020100	registerCountry not exists
01020101	identification is null or length above 30
01020102	buType is null or illegal
01020103	customerIds is null
01020104	customerIds is not exist
01020105	MobileNum is null
01020106	MobileNum is not correct
01020107	NationNum is null
01020108	RandomKey is not correct
01020109	The current verification code has expired.
01020110	Email Send Failed
01020111	modifyType is null
01020112	modifyType is illegal
01020113	addressId is null
01020114	zipcode is null
01020115	country is null
01020116	address is null
01020117	addressId with customerId can not find a address
01020118	Package code is empty
01020119	Software version is empty
01020120	Hardware version is empty

01020121	Software version length can't more than 64
01020122	Device software version format is not correct
01020123	Hardware version length can't more than 64
01020124	The device type does not exist
01020125	Package code does not exist
01020126	The initial state is not release package
01020127	Package code is not initial
01020129	Users binding terminal failure
01020130	Terminal IMEI and terminal password must exist at the same time
01020131	Terminal password is not correct
01020132	The user has been deleted
01020133	Add user failure
01020134	No, IMEI, Initial Package Code, Software Version, Hardware Version, Organization Id, Device Type is not empty
01020135	Terminal password length should not exceed 20
01020136	The device type is not empty
01020137	The operuser type customer does not exist
01020138	User "{0}" have binding terminal, delete failed
01020139	Organization ID is not exist
01020140	Account is null
01020141	Account length can't more than 30
01020142	Account must be email format
01020143	Account password length must be greater than 6, less than 20
01020144	registerSource is null
01020145	password is null
01020146	verification is null
01020147	DefaultOrg is null
01020200	MsgTemplateCode is null
01020201	NationNum is not support
01020202	sms send exceed limits
01020203	local is null
01020204	user not exists
01020205	user has been suspend
01020206	orgId is null
01030001	mvnold is null
01030002	organization is not exist or delete or disable : {0}
01030003	MVNO is not exist
01030004	Organization is empty
01030005	MVNO has customers, can not be deleted
01030006	organization has customers, can not be deleted
01030045	orgState is null or illegal
01030046	orgParentId is null
01030047	orgName not null and length less 30
01030048	orgType is null
01030049	groupType is null
01030050	orgType is not exist
01030051	orgParentId is not exist
01030052	orgParentId is not belong login user
01030053	mvno can not has the same organization name

01030054	orgId is not belong login user
01030055	organization is not belong the mvno
01030056	org is not create by current login Customer
01040001	goodsId is null
01040002	good is not exist : {0}
01040003	upid must not be null
01040004	user package is not exist : {0}
01040005	the package is not belong to user
01040006	package can not cancel
01040007	package expired to cancel
01040008	package can not transfer to self
01040009	srcCustomer and desCustomer not in the same group
01040010	iso2 must not be null
01040011	state must not be null
01040012	state has not defined
01040013	timeFlag has not defined : {0}
01040014	goodsCode must not be null
01040015	goodsType must not be null
01040016	goodsType has not defined : {0}
01040017	goodsPrice must no be null
01040018	categoryId must not be null
01040019	assignCardId must not be null
01040020	pricePlanId must not be null
01040021	good can not add attrs because it has publish
01040022	goodAttrId must not be null
01040023	goodAttr is not exist : {0}
01040024	goodAttr is not belong to good
01040025	operateType must not be null
01040026	goodsType has to be either "DISC" or "BASE"
01040027	good state must be publish
01040028	good already take down, can not publish
01040029	category is not exist
01040030	categoryName must not be null
01040031	categorySName must not be null
01040032	attrCode must not be null
01040033	attrName must not be null
01040034	showFlag must not be null
01040035	langFlag must not be null
01040036	categoryAttrId must not be null
01040037	attrValue must not be null
01040038	attrShowValue must not be null
01040039	langType must not be null
01040040	defaultFlag must not be null
01040041	catagoryAttr is not exist
01040042	categoryAttrValueId must not be null
01040043	categoryAttrValue is not exist
01040044	good effType must be "USER_DEFINED"
01040045	org has configure "BASE" good already, "BASE" good only can configure one

01040046	terminal customer can not configure "BASE" good
01040047	good already exist
01040048	usid must not be null
01040049	custSaleInst is not exist
01040050	custSaleInst List must not be null
01040051	good attr [effType] is not "USER_DEFINED", so effectiveDate and expiryDate input invalid
01040052	good target must be "PERSONAL"
01040053	good target must be "ORG"
01040054	userName and goodsCode is required field, must not be null
01040055	good attr [effType] is "USER_DEFINED", so effectiveDate and expiryDate is required field, must not be null
01040056	attrCode is exist
01050001	attrCode does exist
01050002	productId does not exist
01050003	productCode does exist
01050004	attrId is null
01050005	productId is null
01050006	attrId does not exist
01050007	productCode is null
01050008	serviceType is null
01050009	serviceType does not define
01050010	attrCode does not exists
01050011	productType does not define
01050012	attrCode is null
01050013	productList is null
01050014	PrimGoodsRelaProduct is null
01050015	custAuxProductId is null
01050016	custAuxProductId does not exist
01050017	operateType is null
01050018	operateType does not exist
01070001	amount is null
01070002	payMethod is not support
01070003	orderSource is wrong
01070004	goodsId is null
01070005	Quantity must be greater than zero
01070006	customer username is null
01070007	payType is wrong
01070008	The current customer does not exist
01070009	The accountType is null
01070010	The orderSN is null
01070011	The payStatus is wrong
01070012	The channelType is not support
01070013	The sourceLegalCustomerId is null
01070014	The targetLegalCustomerId is null
01070015	The sourceLegalUsername is null
01070016	The targetLegalUsername is null
01070017	The operatorId is null
01070018	The operatorUsername is null

01070019	The transferType is not support
01070020	The batchName is null
01070021	The maximum length of no more than {0}
01070022	The number is null
01070023	The type is not support
01070024	The batchStatus is not support
01070025	The batchType is not support
01070026	The batchId is null
01070027	The idList is null
01070028	The orgType is not support
01070029	The rechargeCode status is wrong
01070030	The rechargeCode Id is null
01070031	The handleStatus is not support
01070032	The startTime is null
01070033	The endTime is null
01070034	The startTime is greater than the endTime
01070035	The recordSN is null
01070036	The orderId is null
01070037	The orderStatus is not support
01070038	The orderSN is null
01070039	The orderItemId is null
01070040	The deliverState is not support
01080001	terminal not exist,please check the ID
01080002	the IMEI number has existed
01080003	the file not import
01080004	the terminal has been activated,can not delete
01080005	the customer and terminal do not exist
01080006	the customer has been locked,can not activate
01080007	the terminal has been locked,can not activate
01080008	the terminal and customer are activated,no need duplicate
01080009	the terminal has been binded by other customer, are you sure to force to activate?
01080010	the activation object do not exist
01080011	the terminal has been locked,can not deactivate
01080012	the terminal has been locked,no need duplicate
01080013	the terminal has been unlocked,no need duplicate
01080014	the customer has not been activated
01080015	the terminal name can not be null
01080016	the terminal IMEI number can not be null
01080017	the terminal state can not be null
01080018	the terminal code length can not be more than 50
01080019	the terminal name length can not be more than 50
01080020	the IMEI number must be 15 numbers
01080021	the IMEI number can not start with 0
01080022	the terminal password length can not be more than 50
01080023	the seed card IMSI number must be 15 numbers
01080024	the software version length can not be more than 50
01080025	the hardware version length can not be more than 50
01080026	the QR code URL length can not be more than 50

01080027	the terminal description length can not be more than 250
01080028	the terminal type can not be null
01080029	the software version can not be null
01080030	the hardware version can not be null
01080031	the terminalVOList can not be null
01080032	the terminalVOList size can not be above 100
01080033	terminalId or customerId or operatorId is null
01080034	terminal or customer or operator does not exist
01080035	customer or operator's customer type is not right
01080036	the code has existed
01080037	the terminalIds is null
01080038	the terminal list is null
01080100	IMEI associated terminal does not exist
01080101	terminaluserLevel is null
01090001	menu name must not be null
01090002	menu type must not be null
01090003	menu type does not defined
01090004	buttonType must not be null when menu type is "BUTTON"
01090005	menu is not exist : {0}
01090006	menu type not match
01090007	menu id must not be null
01090008	orderno must not be null when menu parentId has value
01090009	menu can not drag to its child nodes
01090010	failed to delete the menu, please delete its all child menus
01090011	menuId List must not be null
01100001	role id must not be null
01100002	role is not exist or delete or disable : {0}
01100003	orgId List must not be null
01100004	system role can not update or delete
01100005	role is in use
01100006	id or requestURI must at least one value
01100007	authority not exist
01100008	role exist
01100009	roleIds is not belong login user
01100013	name is null
01100014	name cannot be more than 50 letters
01100016	comment cannot be more than 200 letters
01100018	enable is null
01100019	roleId List must not be null
01110001	certCode is null
01110002	certCode must be must be email format
01110003	caType is null
01110004	pubKey is null
01110005	priKey is null
01110006	secKey is null
01110007	caInfo not exists
01110008	BSS caInfo not exists
01110009	user certificate is null
01110010	user certificate is error

01110020	email is null
01110021	the format of email is error
01110022	the email has been used
01110023	fileName is null
01110024	pem is null
01110025	amountType is null
01110026	amountType does not exist
01110027	payRelType is null
01110028	relCustomer id is null
01110029	effectiveDate is null
01110030	expiryDate is null
01110031	useMethod is null
01110032	useLimit is null
01110033	useMethod data is error
01110034	payRelType data is error
01110035	objType is null
01110036	objType data is error
01110037	certificate center VerifyToken is null
01110038	CaType not exists
01110039	caVerifyToken not found
01131001	Pricing plan is empty
01131002	Speed gear is not configured
01131003	Finite time type is empty
01131004	Capped amount is empty
01131005	Finite time is empty
01131006	Conditions there is no exist
01131007	Speed gear does not exist
01131008	The speed limit rules ID is empty
01131009	The speed limit rules does not exist
01131010	Speed gear is empty
01131011	Starting flow is empty
01131012	Maximum flow is empty
01131013	Speed gear has been in existence
01131014	Condition name is empty
01131015	Condition ID is empty
01131016	Group ID is empty
01131017	Compare an lvalue is empty
01131018	Compare rvalue is empty
01131019	Comparison operators is empty
01131020	Compare type is empty
01131021	Group number is empty
01131022	You do not have permission to access the interface!
99999999	Fail

3.2.8 Call Entry

For production environment, the start of call entry will be sent to partner offline via email, e.g. <https://saas.ucloudlink.com/bss/>.

3.2.9 Content-Type:application/json

To prevent error "HTTP Status 415" (unsupported media type) during the request, please add 'Content-Type:application/json' in Request-header.

3.2.10 Initial Parameter Provided to Partner

Before partner start integration, uCloudlink will provide initial parameter to partner for the first interface 'Business Customer Login In':

Attribute	Description	Value
partnerCode	Partner code for each MVNO	PARTNERCODE
clientId	Client ID of application of each MVNO	234fe9r90gd03409538kl345
clientSecret	Client Secret of application of each MVNO	234sdfert345345tgrey45345
userCode	Business customer account of each MVNO, one MVNO could have several business customer account if necessary.	USERCODE
password	Business customer account password	password!
mvnoCode	MVNO Code for each MVNO	MVNOCODE

After partner invoke 'Business Customer Login In' interface, partner could get the parameter value of "userId" and "accessToken" which could be used in other interfaces.

3.2.11 Access_token

Access_token need to be attached in every request after business customer had login successfully.

Note: the valid period of access token is only 8 hours after logins, you need to re-login by calling API GrpUserLogin once it expires (result code: 00000007).

These are two method to attach access_token in the request:

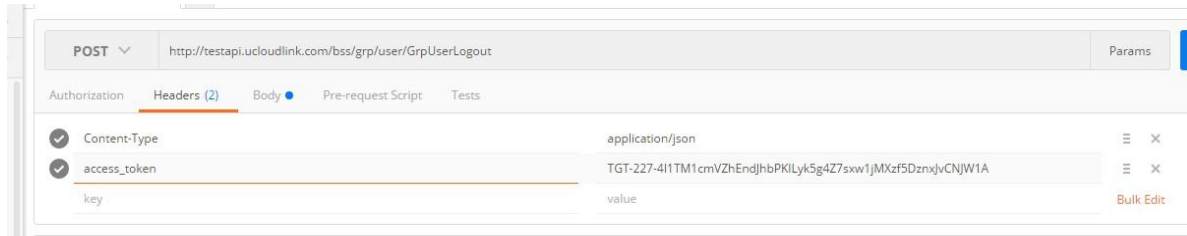
- Attached in URL directly

For example:

https://saas.ucloudlink.com/bss/grp/user/GrpUserLogout?access_token=TGT-227-4I1TM1cmVZhEndJhbPKILyk5g4Z7sXw1jMXzf5DznxJvCNJW1A

If partner invoke the interface via JavaScript in WebSite, then access_token need be attached in URL to avoid website cross-site issue

■ Attached in requestHeader



If partner invoke the interface via Java Process in background, then access_token need to be attached in requestHeader.

3.2.12 Test Devices

Before partner do integration test, partner could get test device IMEI and password from uCloudlink.

Device Type	IMEI	PWD
GlocalMe U2	869680000000000	80800000

3.2.13 Code Demo for invoke interface (JAVA)

```
import java.io.BufferedReader;
import java.io.DataOutputStream;
import java.io.IOException;
import java.io.InputStreamReader;
import java.net.HttpURLConnection;
import java.net.URL;

public class TestApi {

    public static void main(String[] args) throws IOException
    {
        String url =
            "https://saas.ucloudlink.com/bss/grp/user/QueryAccountDetailListInfo?access_token=TGT-256-ck09dxc9MmaaoauJsS49iWdVrMlkRYqdZnhp1phsF9o9MxKbfu";
        URL postUrl = new URL(url);
```

```

        HttpURLConnection connection = (HttpURLConnection)
postUrl.openConnection();
        connection.setDoOutput(true);
        connection.setDoInput(true);
        connection.setRequestMethod("POST");
        connection.setUseCaches(false);
        // set content-type
        connection.setRequestProperty("Content-Type",
"application/json");
        connection.connect();
        DataOutputStream out = new
DataOutputStream(connection.getOutputStream());
        String content = "{" + "\"streamNo\" :
\"XXXXX2016092116475000001\", " + "\"partnerCode\" :
\"XXXXX\", "
                + "\"loginCustomerId\" :
\"57d902ebe85d42329cb23423\", " +
                "\"userCode\" : \"001@XX.com\", "
                + "\"langType\" : \"zh-CN\", " + "
                "\"operateType\" : \"0\", " + "\"currentPage\" : 1, " +
                "\"perPageCount\" : 3"
                + "}";
        System.out.println("====Contents of post
request====");
        System.out.println(content);
        out.writeBytes(content);

        out.flush();
        out.close(); // flush and close
        BufferedReader reader = new BufferedReader(new
InputStreamReader(connection.getInputStream()));
        String line;
        System.out.println("====Contents of post
response====");
        while ((line = reader.readLine()) != null) {
            System.out.println(line);
        }
        reader.close();
        connection.disconnect();
    }

```

```
}
```

3.2.14 Code Demo for invoke interface (PHP)

```
<?php

function http_post_data($url,$data_string){

    $ch=curl_init();
    curl_setopt($ch,CURLOPT_POST,1);
    curl_setopt($ch,CURLOPT_URL,$url);
    curl_setopt($ch,CURLOPT_POSTFIELDS,$data_string);
    curl_setopt($ch,CURLOPT_HTTPHEADER,array(
        'Content-Type: application/json; charset=utf-8',
        'Content-Length: '.strlen($data_string))
    );
    curl_setopt($ch, CURLOPT_SSL_VERIFYPEER, false);
    ob_start();
    curl_exec($ch);
    $return_content=ob_get_contents();
    ob_end_clean();

    $return_code=curl_getinfo($ch,CURLINFO_HTTP_CODE);
    return array($return_code,$return_content);
}

$url="https://saas.ucloudlink.com/bss/grp/user/QueryAccountDetailListInfo?access_token=TGT-256-cK09dxc9MmaaouJsS49iWdVrMlkRYqdZnhp1phsF9o9MxKbfu";

$data='{
    "streamNo": "XXXXXX2016092116475000001",
    "partnerCode": "XXXXXX",
    "loginCustomerId": "57d902ebe342345564cb51d72",
    "userCode": "001@xx.com",
    "langType": "zh-CN",
    "operateType": "0",
    "currentPage": 1,
    "perPageCount": 3
}';

print_r(http_post_data($url,$data));

?>
```

4 Order Mode's Primary Part (Default provide)

These interfaces in this part are common interfaces. If the partners need to test the API interface, we can open these interfaces first. This mode is to purchase the package through the CreatorOrder interface.

4.1 Business customer login in

4.1.1 Overview

Business customer login in then could invoke other interfaces for example: query sub-user list of business customer.

4.1.2 Main Description

API Name	GrpUserLogin
Call Entry	https://saas.ucloudlink.com/bss/grp/noauth/GrpUserLogin
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

4.1.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
langType	Varchar	N	Lang Type	zh-CN; zh-TW; en-US: Default: en-US
clientId	Varchar	Y	Client Id	Eg:bb18c96ca5fd4c82 bca5788f9a0a4311
clientSecret	Varchar	Y	Client Secret	Eg:563b2c3404d0321a 149a0ade25b54c82
userCode	Varchar	Y	User Code	Eg:jingytf_9184@163.c om
password	Varchar	N	Password	Encrypt in MD5 once
mvnoCode	Varchar	Y	MVNO CODE	MVNO CODE

4.1.4 Output Parameter

Attribute	Type	Mandator y	Description	Remarks
-----------	------	---------------	-------------	---------

streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	T	Y	data	Data is in JSON format Eg. "data": { "accessToken": "TGT-454512- GGPNNuteGJLJhRVU LFKXfslaKYJ40eeVv1 BcbSq4O1A1yWxxxx", "userId": "586e0a901ffdba19dd 02xxxx" }

4.2 Business customer login out

4.2.1 Overview

Business customer login in then could invoke other interfaces for example: query sub-user list of business customer.

4.2.2 Main Description

API Name	GrpUserLogout
Call Entry	https://saas.ucloudlink.com/bss/grp/user/GrpUserLogout
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

4.2.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
langType	Varchar	N	Lang Type	zh-CN; zh-TW; en-US; Default: en-US
loginCustomer	Varchar	Y	= userId which	eg:563b2c3404d0321a

Id			returned from API GrpUserLogin	149a0ade25b54c82
accessToken	Varchar	Y	Access Token	eg:TGT-1-0wEFvaKQ7GMcGCBPgVcW6PtJ9TKboK3NB2QyB5A6LHISICMrSp

4.2.4 Output Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP20151113161019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	T	Y	data	Data is in JSON format

4.3 Batch create sub-user of business customer

4.3.1 Overview

Business customer create sub-user of business customer in batch via partner system.

4.3.2 Main Description

API Name	BatchCreateGroupChild
Call Entry	https://saas.ucloudlink.com/bss/grp/user/BatchCreateGroupChild
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

4.3.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
-----------	------	-----------	-------------	---------

streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomerId	Varchar	Y	= userId which returned from API GrpUserLogin	eg:563b2c3404d0321a 149a0ade25b54c82
langType	Varchar	Y	Lang Type	zh-CN zh-TW en-US
childUserVoList	ChildUserVo	Y	Sub-user information	The data is list

4.3.4 ChildUserVO Entity Description

4.3.4.1 Entity Name

Entity Name	ChildUserVO
Description	

4.3.4.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
userCode	Varchar	Y	= the sub-user account which used to bind with a device	Eg:jingytf_9184@163. com
userPwd	Varchar	Y	User Password	Eg:123456
imei	Varchar	N	imei	Eg:356166060755336
tmIPwd	Varchar	N	Terminal Password	Eg:123456
registerType	varchar	Y	Register Type	PHONE; EMAIL
countryCode	varchar	N	Country code	It is mandatory when registerType=PHONE . Such as china is 86
payType	varchar	N		PREPAYMENT POSTPAYMENT
orgCode	varchar	N		It is mandatory when loginCustomerId is top operator

4.3.5 Output Parameter

Attribute	Type	Mandatory	Description	Remarks
-----------	------	-----------	-------------	---------

streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	retData	Y	data	Data is in JSON format

4.3.6 retData Entity Description

4.3.6.1 Entity Name

Entity Name	retData
Description	

4.3.6.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
totalCount	int	Y	Total Count	
successCount	int	Y	Success Page	
errorCount	int	Y	Error Count	
errorList	List	Y	Error List	List: ErrorVO

4.3.7 ErrorVO Entity Description

4.3.7.1 Entity Name

Entity Name	ErrorVO
Description	

4.3.7.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
userCode	Varchar	Y	= the sub-user account which used to bind with a device	Eg:jingytf_9184@163. com
userPwd	Varchar	N	User Password	Eg:123456
imei	Varchar	N	imei	Eg:356166060755336
tmIPwd	Varchar	N	Terminal Password	Eg:123456
errorCode	Varchar	N	Error Code	
errorDesc	Varchar	N	Error Description	

4.4 Query sub-user list of business customer

4.4.1 Overview

Business customer query sub-user list via partner system.

4.4.2 Main Description

API Name	QuerySubUserListInfo
Call Entry	https://saas.ucloudlink.com/bss/grp/user/QuerySubUserListInfo
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

4.4.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomerId	Varchar	Y	= userId which returned from API GrpUserLogin	eg:563b2c3404d0321a 149a0ade25b54c82
langType	Varchar	Y	Lang Type	zh-CN zh-TW en-US
userCode	Varchar	N	= the sub-user account which used to bind with a device	Eg. Abc@163.com .
userCodeQueryType	varchar	N		0: exact query by userCode 1: fuzzy query by userCode
currentPage	int	N	Current Page	default: 1
perPageCount	int	N	Per Page Count	default: 100
deleteFlag	Varchar	N	Delete flag	0: deleted 1: undeleted Default value: 1

4.4.4 Output Parameter

Attribute	Type	Mandatory	Description	Remarks
-----------	------	-----------	-------------	---------

streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	pageData	Y	data	Data is in JSON format

4.4.5 PageData Entity Description

4.4.5.1 Entity Name

Entity Name	pageData
Description	

4.4.5.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
perPageCount	int	Y	Per Page Count	
currentPage	int	Y	Current Page	
totalCount	int	Y	Total Count	
totalPageCount	int	Y	Total Page Count	
dataList	List	Y	Data List	List: SubUserVO

4.4.6 SubUserVO Entity Description

4.4.6.1 Entity Name

Entity Name	SubUserVO
Description	

4.4.6.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
userCode	Varchar	Y	= the sub-user account which used to bind with a device	Eg:jingytf_9184@163. com
customerId	Varchar	Y	Customer Id	Eg:563b2c3404d0321 a149a0ade25b54c82
imei	Varchar	N	imei	Eg:356166060755336
tmIPwd	Varchar	N	Terminal Password	Eg:123456
status	Varchar	N	Customer status	enable: customer is enable; disable: customer is disable

balance	Varchar	N	Unit: cents	Eg. 100 cents (= 1 RMB)
hVersion	Varchar	N	Hardware Version	
sVersion	Varchar	N	Software Version	
currencyType	Varchar	N	Currency Type	eg: RMB,EUR
lockFlag	Varchar	N	Lock Flag	0: locked; 1:unlocked
activationFlag	Varchar	N	Activation Flag	0: activated; 1: un-activated;
createDate	Varchar	N	Create Date	
remark	Varchar	N	remark	

4.5 Query available offer list for sale

4.5.1 Overview

Business customer query available offer list which can be ordered via partner system for sub-users of business customer.

4.5.2 Main Description

API Name	QueryGrpOfferList
Call Entry	https://saas.ucloudlink.com/bss/grp/goods/QueryGrpOfferList
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

4.5.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomerId	Varchar	Y	= userId which returned from API GrpUserLogin	eg:563b2c3404d0321a 149a0ade25b54c82
langType	Varchar	Y	Lang Type	zh-CN zh-TW en-US
bgTime	long	N	Begin Time	Timestamp in GMT0
endTime	long	N	End Time	Timestamp in GMT0
currentPage	int	N	Current Page	Default:1

perPageCount	int	N	Per Page Count	Default:100
channelType	Varchar	N	Channel Type	GRP
mcc	Varchar	N	mcc	CHINA=CN
mccFlag	Varchar	N	Mcc Flag	WHITE: white list BLACK: Black list
goodsCode	Varchar	N	Goods Code	
categoryCode	Varchar	Y	Category Code	LLTC: Data bundle; BTTC: Daily bundle; SWBY: Monthly bundle;
goodsType	varchar	N	Goods Type	DISC: Daily/Monthly bundle; PKAG: Data bundle;

4.5.4 Output Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	pageData	Y	data	Data is in JSON format

4.5.5 PageData Entity Description

4.5.5.1 Entity Name

Entity Name	pageData
Description	

4.5.5.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
perPageCount	int	Y	Per Page Count	
currentPage	int	Y	Current Page	
totalCount	int	Y	Total Count	
totalPageCoun	int	Y	Total Page	

t			Count	
dataList	List	Y	Data List	List: GoodsVO

4.5.6 GoodsVO Entity Description

4.5.6.1 Entity Name

Entity Name	GoodsVO
Description	

4.5.6.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
goodsId	Varchar	Y	Goods Id	
goodsCode	Varchar	Y	Goods Code	
goodsName	Varchar	Y	Goods Name	
goodsPrice	double	Y	Goods Price	
status	Varchar	Y	status	0: initial; 1: release; 2: offline
mccList	List	N	Mcc List	["CN", "HK"]
mccFlag	Varchar	N	Mcc Flag	WHITE: white list BLACK: Black list
createTime	long	Y	Create Time	Timestamp in GMT0, eg:1447225157471
flowByte	double	Y	Flow Byte	MB eg:0.01MB
period	Varchar	N	period	Day eg: 2days
currencyType	Varchar	Y	Currency Type	EUR RMB
categoryCode	Varchar	N	Category Code	LLTC: Data bundle; BTTC: Daily bundle; SWBY: Monthly bundle;
periodUnit	varchar	N	Period unit	DAY; MONTH
attrMap	map	Y	Extra Attributes	{ "pkDesc": " sellsGood_zjm_0800", "effType": "NEXT_MONTH", "title": "sellsGood_zjm_0800" , "timeZone": "GMT+8", "period": "30", "periodUnit": "MONTH" }

4.5.6.3 AttrMap

Attribute	Type	Description	Remarks
-----------	------	-------------	---------

effType	Effect type		USER_DEFINED: user-defined the effective time and the expire Time. IMMEDIATELY : Immediately effect. MORROW: The next data effect. NEXT_MONTH: next month effect. ACTIVATION: effect when the offer activated
period	number of period		
periodUnit	Period unit		MONTH, DAY
timeZone	timezone		eg: GMT+0
title	title		
pkDesc	description		

4.6 Create order for sub-user

4.6.1 Overview

Business customer create order for sub-user to set offer via partner system.

4.6.2 Main Description

API Name	GrpCreateOrder
Call Entry	https://saas.ucloudlink.com/bss/grp/order/GrpCreateOrder
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

4.6.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP20151113161019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomer Id	Varchar	Y	= userId which returned from API GrpUserLogin	eg:563b2c3404d0321a149a0ade25b54c82
userCode	Varchar	Y	= the sub-user	eg:jingytf_1984@163.c

			account which used to bind with a device	om
channelType	Varchar	Y	Channel Type	GRP
orderType	Varchar	Y	Order Type	BUYPKG
goodsList	List	Y	Goods List	OrderItemVo
orderMark	Varchar	N	Order Mark	
currencyType	Varchar	Y	Currency Type	
payMethod	Varchar	Y	Pay Method	ACCOUNT_AMOUNT: Pay by balance

4.6.4 OrderItemVO Entity Description

4.6.4.1 Entity Name

Entity Name	OrderItemVO
Description	

4.6.4.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
goodsId	Varchar	Y	Goods Id	
quantity	int	Y	quantity	
effectiveTime	long	N	Effective Time	Timestamp in GMT0, eg: 1447225157471 It is mandatory only when Effective type of packages is user-defined.
expireTime	long	N	Expire Time	Timestamp in GMT0, eg: 1447225157471 It is mandatory only when Effective type of packages is user-defined.
flowSize	int	N	Flow size	Eg.flowSzie Unit: MB
mccList	List	N	mccList	Eg: ["CN","JP","HK"]

4.6.5 Output Parameter

Attribute	Type	Mandator y	Description	Remarks
-----------	------	---------------	-------------	---------

streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	T	N	data	{"orderSN":"5678c8bb 26cf673c833ef86b","a mount":12.03,"currenc yType":"EUR","orderId ":"5678c8bb26cf673c8 33oi987"}

4.7 Query offer list of sub-user

4.7.1 Overview

Business customer query offer list of sub-user via partner system.

4.7.2 Main Description

API Name	QueryUserOfferList
Call Entry	https://saas.ucloudlink.com/bss/grp/goods/QueryUserOfferList
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

4.7.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomerId	Varchar	Y	= userId which returned from API GrpUserLogin	eg:563b2c3404d0321a 149a0ade25b54c82
langType	Varchar	Y	Lang Type	zh-CN zh-TW en-US
userCode	Varchar	Y	= the sub-user account which used to bind	

			with a device	
bgTime	long	N	Begin Time	Timestamp in GMT0
endTime	long	N	End Time	Timestamp in GMT0
currentPage	int	N	Current Page	Default:1
perPageCount	int	N	Per Page Count	Default:100
flag	Varchar	Y	Flag	0:using, not active, valid; 1: used out, expired, invalid 2: for all status
categoryCode	Varchar	N	Category Code	LLTC: Data bundle; BTTC: Daily bundle; SWBY: Monthly bundle
iso2	Varchar	N	iso2	Eg: CN
goodsType	varchar	Y	Goods Type	DISC: Daily/Monthly bundle, include BTTC, SWBY etc; PKAG: Data bundle; ALL: The above mentioned;
orderId	varchar	N	= orderId which returned from API GrpCreateOrder	

4.7.4 Output Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP20151113161019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	pageData	Y	data	Data is in JSON format

4.7.5 PageData Entity Description

4.7.5.1 Entity Name

Entity Name	pageData
Description	

4.7.5.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
perPageCount	int	Y	Per Page Count	
currentPage	int	Y	Current Page	
totalCount	int	Y	Total Count	
totalPageCount	int	Y	Total Page Count	
dataList	List	Y	Data List	List: OrderRelationVo

4.7.6 OrderRelationVO Entity Description

4.7.6.1 Entity Name

Entity Name	OrderRelationVO
Description	

4.7.6.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
relationId	Varchar	Y	OrderRelationId	
customerId	Varchar	Y	Customer Id	
customerName	Varchar	Y	Customer Name	
goodsId	Varchar	Y	Goods Id	
goodsName	Varchar	Y	Goods Name	
goodsCode	Varchar	Y	Goods Code	
status	Varchar	Y	Status	1.Status for goodsType =PKAG: - NOT_ACTIVATED - IN_USING - USE_END - EXPIRE - TRANSFER - UNSUBSCRIBE 2.Status for goodsType=DISC: - NOT_ACTIVATED - VALID - INVALID
mccList	List	N	Mcc List	["CN","HK"]
mccFlag	Varchar	N	Mcc Flag	WHITE: white list BLACK: Black list
createTime	long	Y	Create Time	Timestamp in GMT0,

				eg:1447225157471
effectiveTime	long	Y	Effective Time	Timestamp in GMT0, eg: 1447225157471
expiryTime	long	Y	Expiry Time	Timestamp in GMT0, eg:1447225157471
flowByte	double	Y	Total Flow Byte For goodsType=PKA G	MB, eg:1024MB
accumulatedFlowList	List	Y	accumulatedFlowList For goodsType=DISC	cycle – unit of accumulated cycle which 0: hour; 1: day; 2: month; 3: year cycleValue – number of the cycle units eg. "accumulatedFlowList" : { "accumulatedFlow": 0.01, "limitFlag": false, "limitLevel": 0, "cycle": 1, "cycleValue": 1, "pricePlan": "xxxx53952a", "rateUp": null, "rateDown": null, "mccList": [], "noMccList": [], "countryList": ["VU", "VN", "LK"], "totalFlow": 1024 }]
surplusFlowbyte	double	Y	Data balance For goodsType=PKA G	MB eg:305.38MB
period	Varchar	N	period	Day eg: 2days
categoryCode	Varchar	Y	Category Code	LLTC: Data bundle; BTTC: Daily bundle; SWBY: Monthly bundle; ALL: for all type
goodsType	varchar	N	Goods Type	DISC: Daily/Monthly bundle; PKAG: Data bundle; ALL: The above mentioned;
periodUnit	varchar	N	Period unit	DAY; MONTH

attrMap	map	Y	Extra Attributes	{ "pkDesc": " sellsGood_zjm_0800", "effType": "NEXT_MONTH", "title": "sellsGood_zjm_0800" , "timeZone": "GMT+8", "period": "30", "periodUnit": "MONTH" }
orderId	varchar	Y	Order ID	
terminalType	varchar	Y	Terminal Type	

4.7.6.3 AttrMap Attribute

Attribute	Type	Description	Remarks
effType	Effect type		USER_DEFINED: user-defined the effective time and the expire Time. IMMEDIATELY : Immediately effect. MORROW: The next data effect. NEXT_MONTH: next month effect. ACTIVATION: effect when the offer activated
period	number of period		
periodUnit	Period unit		MONTH, DAY
timeZone	timezone		eg: GMT+0
title	title		
pkDesc	description		

5 Order Mode's Secondary Part (Open only after application)

This part of the interface is selectively open according to the customer's specific business scenario.

5.1 Stop and start sub-user (batch)

5.1.1 Overview

Used to disable/enable userCode. Mi-fi service will be interrupted once userCode is disable. And customer should reboot the device if userCode enabled again.

5.1.2 Main Description

API Name	StopSubUser
Call Entry	https://saas.ucloudlink.com/bss/grp/user/StopSubUser
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

5.1.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomerId	Varchar	Y	= userId which returned from API GrpUserLogin	eg:563b2c3404d0321a 149a0ade25b54c82
langType	Varchar	Y	Lang Type	zh-CN zh-TW en-US
operateType	Varchar	Y	Operate Type	0:stop (Disable) 1:start (Enable)
userCodeList	List	Y	User Code List	= the sub-user account which used to bind with a device

5.1.4 Output Parameter

Attribute	Type	Mandator y	Description	Remarks
-----------	------	---------------	-------------	---------

streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	retData	Y	data	Data is in JSON format

5.1.5 retData Entity Description

5.1.5.1 Entity Name

Entity Name	retData
Description	

5.1.5.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
totalCount	int	Y	Total Count	
successCount	int	Y	Success Page	
errorCount	int	Y	Error Count	
errorList	List	Y	Error List	List: ErrorVO

5.1.6 ErrorVO Entity Description

5.1.6.1 Entity Name

Entity Name	ErrorVO
Description	

5.1.6.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
userCode	Varchar	Y	= the sub-user account which used to bind with a device	Eg:jingytf_9184@163. com
errorCode	Varchar	N	Error Code	
errorDesc	Varchar	N	Error Description	

5.2 Delete and resume sub-user (batch)

5.2.1 Overview

Business customer delete and resume sub-user of business customer in batch via partner system.

5.2.2 Main Description

API Name	DeleteSubUser
Call Entry	https://saas.ucloudlink.com/bss/grp/user/DeleteSubUser
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

5.2.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomerId	Varchar	Y	= userId which returned from API GrpUserLogin	eg:563b2c3404d0321a 149a0ade25b54c82
langType	Varchar	Y	Lang Type	zh-CN zh-TW en-US
operateType	Varchar	Y	Operate Type	0:delete,1:resume
userCodeList	List	Y	User Code List	= the sub-user account which used to bind with a device

5.2.4 Output Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	retData	Y	data	Data is in JSON format

5.2.5 retData Entity Description

5.2.5.1 Entity Name

Entity Name	retData
Description	

5.2.5.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
totalCount	int	Y	Total Count	
successCount	int	Y	Success Page	
errorCount	int	Y	Error Count	
errorList	List	Y	Error List	List: ErrorVO

5.2.6 ErrorVO Entity Description

5.2.6.1 Entity Name

Entity Name	ErrorVO
Description	

5.2.6.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
userCode	Varchar	Y	= the sub-user account which used to bind with a device	Eg:jingytf_9184@163.com
errorCode	Varchar	N	Error Code	
errorDesc	Varchar	N	Error Description	

5.3 Reset sub-user password

5.3.1 Overview

Business customer reset sub-user password of business customer via partner system. The password is used for device login in.

5.3.2 Main Description

API Name	ResetPwdForSubUser
Call Entry	https://saas.ucloudlink.com/bss/grp/user/ResetPwdForSubUser
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

5.3.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomerId	Varchar	Y	= userId which returned from API GrpUserLogin	eg:563b2c3404d0321a 149a0ade25b54c82
langType	Varchar	Y	Lang Type	zh-CN zh-TW en-US
userCode	Varchar	Y	= the sub-user account which used to bind with a device	Eg:jingytf_9184@163.c om
password	Varchar	Y	New Password	Should input the new password which encrypted in MD5.

5.3.4 Output Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	T	Y	data	Data is in JSON format

5.4 Bind device with sub-user

5.4.1 Overview

Business customer bind device with sub-user of business customer via partner system.

5.4.2 Main Description

API Name	BindDevice
Call Entry	https://saas.ucloudlink.com/bss/grp/tml/BindDevice
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

5.4.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomerId	Varchar	Y	= userId which returned from API GrpUserLogin	eg:563b2c3404d0321a 149a0ade25b54c82
langType	Varchar	Y	Lang Type	zh-CN zh-TW en-US
userCode	Varchar	Y	= the sub-user account which used to bind with a device	
imei	Varchar	Y	imei	Eg:356166060755336
tmlPwd	Varchar	Y	Tml Pwd	
forceFlag	boolean	Y	Force Flag	true: bind in force even the device is binding by other sub-user. false: do not bind if the device is binding by other sub-user.

5.4.4 Output Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result	

			Description	
data	T	Y	data	Data is in JSON format

5.5 Unbind device with sub-user (batch)

5.5.1 Overview

Business customer unbind device with sub-user of business customer in batch via partner system.

5.5.2 Main Description

API Name	BatchUnBindDevice
Call Entry	https://saas.ucloudlink.com/bss/grp/tml/BatchUnBindDevice
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

5.5.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomer Id	Varchar	Y	= userId which returned from API GrpUserLogin	eg:563b2c3404d0321a 149a0ade25b54c82
langType	Varchar	Y	Lang Type	zh-CN zh-TW en-US
imeiList	List	Y	IMEI List	

5.5.4 Output Parameter

Attribute	Type	Mandator y	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000

resultDesc	Varchar	Y	Result Description	
data	retData	Y	data	Data is in JSON format

5.5.5 retData Entity Description

5.5.5.1 Entity Name

Entity Name	retData
Description	

5.5.5.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
totalCount	int	Y	Total Count	
successCount	int	Y	Success Page	
errorCount	int	Y	Error Count	
errorList	List	Y	Error List	List: ErrorVO

5.5.6 ErrorVO Entity Description

5.5.6.1 Entity Name

Entity Name	ErrorVO
Description	

5.5.6.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
imei	Varchar	Y	imei	
errorCode	Varchar	N	Error Code	
errorDesc	Varchar	N	Error Description	

5.6 Query device binding Information

5.6.1 Overview

Business customer query sub-user device binding information via partner system.

5.6.2 Main Description

API Name	QueryBindingRelationInfo
Call Entry	https://saas.ucloudlink.com/bss/grp/tml/QueryBindingRelationInfo
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

5.6.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomerId	Varchar	Y	= userId which returned from API GrpUserLogin	eg:563b2c3404d0321a 149a0ade25b54c82
langType	Varchar	Y	Lang Type	zh-CN zh-TW en-US
userCode	Varchar	N	= the sub-user account which used to bind with a device	Eg:jingytf_1984@163.c om. At least userCode or imei is mandatory. If both userCode and imei are input, then only userCode is used to query.
imei	Varchar	N	imei	Eg:356166060755336. At least userCode or imei is mandatory.If both userCode and imei are input, then only userCode is used to query.

5.6.4 Output Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	TerminalActivationVo	Y	data	Data is in JSON format

5.6.5 TerminalActivationVo Entity Description

5.6.5.1 Entity Name

Entity Name	TerminalActivationVo
Description	

5.6.5.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
id	Varchar	Y	id	
customerId	Varchar	Y	Customer Id	
customerName	Varchar	N	Customer Name	
imei	Varchar	Y	imei	Eg:356166060755336
createTime	long	Y	Create Time	Timestamp for create time in GMT0, eg:1447225157471
status	Varchar	N	status	BINDING: userCode is binded with the device, but not use mi-fi service yet. BINDED: userCode is binded with the device, and it is using/was used mi-fi service on the device.
isLocked	boolean	N	Is Locked	true: locked; false: unlocked
terminalType	Varchar	N	Terminal Type	Eg. G2,E1,...

5.7 Lock and unlock device (batch)

5.7.1 Overview

Business customer lock and unlock device with sub-user of business customer in batch via partner system.

The device can't be bound with another sub-user if it is locked by a sub-user. And it can be unlocked.

The device can't be locked/unlocked if there is no sub-user binding.

5.7.2 Main Description

API Name	LockDevice
Call Entry	https://saas.ucloudlink.com/bss/grp/tml/LockDevice
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

5.7.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomerId	Varchar	Y	= userId which returned from API GrpUserLogin	eg:563b2c3404d0321a 149a0ade25b54c82
langType	Varchar	Y	Lang Type	zh-CN zh-TW en-US
operateType	Varchar	Y	Operate Type	0:lock,1:unlock
imeiList	List	Y	IMEI List	

5.7.4 Output Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	retData	Y	data	Data is in JSON format

5.7.5 retData Entity Description

5.7.5.1 Entity Name

Entity Name	retData
Description	

5.7.5.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
totalCount	int	Y	Total Count	
successCount	int	Y	Success Page	
errorCount	int	Y	Error Count	
errorList	List	Y	Error List	List: ErrorVO

5.7.6 ErrorVO Entity Description

5.7.6.1 Entity Name

Entity Name	ErrorVO
Description	

5.7.6.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
imei	Varchar	Y	imei	
errorCode	Varchar	N	Error Code	
errorDesc	Varchar	N	Error Description	

5.8 Top up for sub-user

5.8.1 Overview

Business customer top up for sub-user to set prepaid customer balance via partner system.

5.8.2 Main Description

API Name	TopUpForSubUser
Call Entry	https://saas.ucloudlink.com/bss/grp/topup/TopUpForSubUser
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

5.8.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomerId	Varchar	Y	= userId which returned from API GrpUserLogin	eg:563b2c3404d0321a 149a0ade25b54c82
langType	Varchar	N	langType	
operateType	Varchar	Y	operateType	0: top up via voucher; 1: top up via business customer account
userCode	Varchar	Y	= the sub-user account which used to bind	Eg:jingytf_9184@163.com

			with a device	
voucherCode	Varchar	N	voucherCode	It's mandatory when top up via voucher
amount	double	N	amount	Eg. 100 Unit of RMB is fen; Unit of dollar is cent

5.8.4 Output Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	T	N	data	BalanceVo

5.8.5 BalanceVO Entity Description

5.8.5.1 Entity Name

Entity Name	BalanceVO
Description	

5.8.5.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
userCode	Varchar	N	= the sub-user account which used to bind with a device	
customerId	Varchar	N	customerId	
accountId	Varchar	N	accountId	
balance	double	N	balance	Sub-user account balance after top up

5.9 Query top up list of sub-user

5.9.1 Overview

Business customer query top up list of sub-user via partner system.

5.9.2 Main Description

API Name	QueryTopUpListInfo
Call Entry	https://saas.ucloudlink.com/bss/grp/topup/QueryTopUpListInfo
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

5.9.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomerId	Varchar	Y	= userId which returned from API GrpUserLogin	eg:563b2c3404d0321a 149a0ade25b54c82
langType	Varchar	Y	Lang Type	zh-CN zh-TW en-US
userCode	Varchar	Y	= the sub-user account which used to bind with a device	
bgTime	long	N	Begin Time	Timestamp in GMT0
endTime	long	N	End Time	Timestamp in GMT0
currentPage	int	N	Current Page	Default:1
perPageCount	int	N	Per Page Count	Default:100

5.9.4 Output Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	pageData	Y	data	Data is in JSON format

5.9.5 PageData Entity Description

5.9.5.1 Entity Name

Entity Name	pageData
Description	

5.9.5.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
perPageCount	int	Y	Per Page Count	
currentPage	int	Y	Current Page	
totalCount	int	Y	Total Count	
totalPageCount	int	Y	Total Page Count	
dataList	List	Y	Data List	List: BalanceChargeVo

5.9.6 BalanceChargeVO Entity Description

5.9.6.1 Entity Name

Entity Name	BalanceChargeVO
Description	

5.9.6.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
operateType	Varchar	Y	Operate Type	PROXY_RECHARGE: top up by business customer account;
amount	double	Y	amount	
userCode	Varchar	Y	= the sub-user account which used to bind with a device	
customerId	Varchar	Y	Customer Id	
createTime	long	Y	Create Time	Timestamp in GMT0
operateUserCode	Varchar	Y	Operate User Code	
channelType	Varchar	Y	Channel Type	eg: WEB,APP,SAAS

5.10 Query account detail of sub-user

5.10.1 Overview

Business customer query account detail of sub-user via partner system.

5.10.2 Main Description

API Name	QueryAccountDetailListInfo
Call Entry	https://saas.ucloudlink.com/bss/grp/user/QueryAccountDetailListInfo
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

5.10.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomerId	Varchar	Y	= userId which returned from API GrpUserLogin	eg:563b2c3404d0321a 149a0ade25b54c82
langType	Varchar	Y	Lang Type	zh-CN zh-TW en-US
operateType	varchar	Y	Operate Type	0:query business customer account; 1:query sub-user account
userCode	Varchar	Y	= the sub-user account which used to bind with a device	
bgTime	long	N	Begin Time	Timestamp in GMT0
endTime	long	N	End Time	Timestamp in GMT0
currentPage	int	N	Current Page	Default:1
perPageCount	int	N	Per Page Count	Default:100

5.10.4 Output Parameter

Attribute	Type	Mandatory	Description	Remarks
-----------	------	-----------	-------------	---------

streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	pageData	Y	data	Data is in JSON format

5.10.5 PageData Entity Description

5.10.5.1 Entity Name

Entity Name	pageData
Description	

5.10.5.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
perPageCount	int	Y	Per Page Count	
currentPage	int	Y	Current Page	
totalCount	int	Y	Total Count	
totalPageCount	int	Y	Total Page Count	
dataList	List	Y	Data List	List: BalanceDetailVO

5.10.6 BalanceDetailVO Entity Description

5.10.6.1 Entity Name

Entity Name	BalanceDetailVO
Description	

5.10.6.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
userCode	Varchar	Y	= the sub-user account which used to bind with a device	
imei	Varchar	Y	imei	imei
mainType	varchar	Y	Main Type	ADD,SUBSTRACT
opType	Varchar	Y	Operate Type	AMOUNT_CHARGE: PAYG charging; PROXY_RECHARGE: top up via business customer account;

				TOPUP_RECHARGE: top up via voucher; ACCOUNT_PURCHASE_PACKAGE: purchase offer; TRANSFER_AMOUNT: T: transfer balance between business account
amount	double	Y	amount	
createTime	long	N	Create Time	
goodsNames	List	N	Goods Names	Eg:["UK 500M","CH 500M"]
currencyType	varchar	N	Currency Type	Eg:EUR

5.11 Batch query sub-user list of business customer

5.11.1 Overview

Business customer query sub-user list in batch via partner system.

5.11.2 Main Description

API Name	QuerySubUserListByBatch
Call Entry	https://saas.ucloudlink.com/bss/grp/user/QuerySubUserListByBatch
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

5.11.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomerId	Varchar	Y	= userId which returned from API GrpUserLogin	eg:563b2c3404d0321a 149a0ade25b54c82
langType	Varchar	Y	Lang Type	zh-CN zh-TW en-US
operateType	varchar	Y	Operate Type	0:query sub-user by

				imei; 1:query sub-user by userCode;
userCodeList	List	N	User Code List	It is mandatory when operateType is 1.
imeiList	List	N	Imei List	It is mandatory when operateType is 0.

5.11.4 Output Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	pageData	Y	data	Data is in JSON format

5.11.5 QuerySubUserListByBatchVO Entity Description

5.11.5.1 Entity Name

Entity Name	QuerySubUserListByBatchVO
Description	

5.11.5.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
userCode	Varchar	Y	= the sub-user account which used to bind with a device	Eg:jingytf_9184@163. com
customerId	Varchar	Y	Customer Id	Eg:563b2c3404d0321 a149a0ade25b54c82
imei	Varchar	N	imei	Eg:356166060755336
tmIPwd	Varchar	N	Terminal Password	Eg:123456
status	Varchar	N	Customer status	enable: customer is enable; disable: customer is disable
balance	Varchar	N	balance	
hVersion	Varchar	N	Hardware Version	
sVersion	Varchar	N	Software Version	

currencyType	Varchar	N	Currency Type	eg: RMB
lockFlag	Varchar	N	Lock Flag	0: locked; 1:unlocked
activationFlag	Varchar	N	Activation Flag	0: activated; 1: un-activated;
createDate	Varchar	N	Create Date	
remark	Varchar	N	remark	

5.12 Operate offer for sub-user

5.12.1 Overview

Note: this interface is only used to synchronize orderRelation from Partner's system to uCloudlink's system but not generate orders.

5.12.2 Main Description

API Name	SyncOrderRelationFromUP
Call Entry	https://saas.ucloudlink.com/bss/grp/goods/SyncOrderRelationFromUP
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

5.12.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomerId	Varchar	Y	= userId which returned from API GrpUserLogin	eg:563b2c3404d0321a 149a0ade25b54c82
langType	Varchar	Y	Lang Type	zh-CN zh-TW en-US
userCode	Varchar	N	= the sub-user account which used to bind with a device	Either userCode or imei needs input value. If input value for both userCode and IMEI, then userCode will be the main.
imei	Varchar	N	IMEI	Pre-condition: this IMEI has been bound with a userCode. eg:356166060755336
operateType	Varchar	Y	Operate Type	0: Add orderRelation and synchronize to

				uCloudlink. 1: Cancel orderRelation and synchronize to uCloudlink. 2: Update orderRelation and synchronize to uCloudlink.
goodsVoList	goodsVo	Y	GoodsVo	

5.12.4 GoodsVo Entity Description

5.12.4.1 Entity Name

Entity Name	GoodsVo
Description	

5.12.4.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
orderRelationId	Varchar	N	orderRelationId= relationId which return from QueryUserOffer List	It is mandatory when operateType=1 or operateType=2.
goodsCode	Varchar	N	Goods Code	It is mandatory when operateType=0
goodsType	Varchar	N	Goods Type	It is mandatory when operateType=1 or operateType=2. DISC: Daily/Monthly bundle; PKAG: Data bundle;
effectiveDate	Long	N	Effective Time	It is mandatory when operateType is 0 or 2. Timestamp in GMT0, eg: 1447225157471
expiryDate	Long	N	Expire Time	It is mandatory when operateType is 0 or 2. Timestamp in GMT0, eg: 1447225157471

5.12.5 Output Parameter

Attribute	Type	Mandatory	Description	Remarks
-----------	------	-----------	-------------	---------

streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	T	Y	data	Data is in JSON format

5.13 Query offer list of sub-user by orderId

5.13.1 Overview

Business customer to query offer list of sub-user by order-id via partner system.

5.13.2 Main Description

API Name	QueryOrderRelationByOrderId
Call Entry	https://saas.ucloudlink.com/bss/grp/order/QueryOrderRelationByOrderId
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

5.13.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomerId	Varchar	Y	= userId which returned from API GrpUserLogin	eg:563b2c3404d0321a 149a0ade25b54c82
langType	Varchar	Y	Lang Type	zh-CN zh-TW en-US
orderId	varchar	Y	Order Id	From API

5.13.4 Output Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	T	Y	data	List: OrderRelationSubVo

5.13.5 OrderRelationSubVO Entity Description

5.13.5.1 Entity Name

Entity Name	OrderRelationSubVO
Description	

5.13.5.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
customerId	Varchar	Y	Customer Id	
userCode	Varchar	Y	= the sub-user account which used to bind with a device	
goodsId	Varchar	Y	Goods Id	
goodsName	Varchar	Y	Goods Name	
status	Varchar	Y	status	1.Status for goodsType =PKAG: - NOT_ACTIVATED - IN_USING - USE_END - EXPIRE - TRANSFER - UNSUBSCRIBE 2.Status for goodsType=DISC: - NOT_ACTIVATED - VALID - INVALID
orderRelationId	varchar	Y	Order Relation Id	

5.14 Unsubscribe order for sub-user

5.14.1 Overview

Business customer to unsubscribe order for sub-user.

5.14.2 Main Description

API Name	CancelOrder
Call Entry	https://saas.ucloudlink.com//bss/grp/order/CancelOrder
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

5.14.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomerId	Varchar	Y	Login Customer Id	eg:563b2c3404d0321a 149a0ade25b54c82
langType	Varchar	Y	Lang Type	zh-CN zh-TW en-US
orderId	varchar	N	Order id which returned from API GrpCreateOrder	Either orderId or orderSN needs input value.orderId And orderSNchoice, orderId priority
orderSN	varchar	N	Order SN which returned from GrpCreateOrder	

5.14.4 Output Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	pageData	Y	data	Data is in JSON format

5.15 Operate OrderRelation for sub-user

5.15.1 Overview

Note: this interface is only used to synchronize orderRelation from Partner's system to uCloudlink's system but not generate orders.

5.15.2 Main Description

API Name	OperateOrderRelation
Call Entry	https://saas.ucloudlink.com/bss/grp/goods/OperateOrderRelation
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

5.15.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP20151113161019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomerId	Varchar	Y	= userId which returned from API GrpUserLogin	eg:563b2c3404d0321a149a0ade25b54c82
langType	Varchar	Y	Lang Type	zh-CN zh-TW en-US
userCode	Varchar	N	= the sub-user account which used to bind with a device	Either userCode or imei needs input value. If input value for both userCode and IMEI, then userCode will be the main.
imei	Varchar	N	IMEI	Pre-condition: this IMEI has been bound with a userCode. eg:356166060755336
goodsVoList	goodsVo	Y	GoodsVo	

5.15.4 GoodsVo Entity Description

5.15.4.1 Entity Name

Entity Name	GoodsVo
Description	

5.15.4.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
goodsCode	Varchar	Y	Goods Code	It is mandatory
effectiveDate	Long	Y	Effective Time	It is mandatory. Timestamp in GMT0, eg: 1447225157471
expiryDate	Long	Y	Expire Time	It is mandatory. Timestamp in GMT0, eg: 1447225157471
flowSize	int	N	Flow size	Eg.flowSzie Unit: MB
mccList	List	N	mccList	Eg: ["CN","JP","HK"]

5.15.5 Output Parameter

Attribute	Type	Mandator y	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	T	Y	data	Data is in JSON format

5.16 Disable OrderRelation for sub-user

5.16.1 Overview

Note: this interface is only used to invalid offer from Partner's system to uCloudlink's system but not generate orders.

5.16.2 Main Description

API Name	DisableOrderRelation
Call Entry	https://saas.ucloudlink.com/bss/grp/goods/DisableOrderRelation
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

5.16.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
-----------	------	-----------	-------------	---------

streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomerId	Varchar	Y	= userId which returned from API GrpUserLogin	eg:563b2c3404d0321a 149a0ade25b54c82
langType	Varchar	Y	Lang Type	zh-CN zh-TW en-US
userCode	Varchar	N	= the sub-user account which used to bind with a device	Either userCode or imei needs input value. If input value for both userCode and IMEI, then userCode will be the main.
imei	Varchar	N	IMEI	Pre-condition: this IMEI has been bound with a userCode. eg:356166060755336
goodsVoList	goodsVo	Y	GoodsVo	

5.16.4 GoodsVo Entity Description

5.16.4.1 Entity Name

Entity Name	GoodsVo
Description	

5.16.4.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
orderRelationId	Varchar	N	RelationId	Either orderRelationId or goodsCode needs input value. If input value for both orderRelationId and goodsCode, then orderRelationId will be the main
goodsCode	Varchar	N	Goods Code	

5.16.5 Output Parameter

Attribute	Type	Mandatory	Description	Remarks
-----------	------	-----------	-------------	---------

streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	T	Y	data	Data is in JSON format

5.17 Update OrderRelation for sub-user

5.17.1 Overview

Note: this interface is only used to update orderRelation from Partner's system to uCloudlink's system but not generate orders.

5.17.2 Main Description

API Name	UpdateOrderRelation
Call Entry	https://saas.ucloudlink.com/bss/grp/goods/UpdateOrderRelation
HTTP Request Mode	HTTP-POST
Format	JSON
API Authorization	Yes
Version	V1.0

5.17.3 Input Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP2015111316 1019000001
partnerCode	Varchar	Y	Partner Code	
loginCustomerId	Varchar	Y	= userId which returned from API GrpUserLogin	eg:563b2c3404d0321a 149a0ade25b54c82
langType	Varchar	Y	Lang Type	zh-CN zh-TW en-US
userCode	Varchar	N	= the sub-user account which used to bind with a device	Either userCode or imei needs input value. If input value for both userCode and IMEI, then userCode will be the main.
imei	Varchar	N	IMEI	Pre-condition: this IMEI has been bound

				with a userCode. eg:356166060755336
goodsVoList	goodsVo	Y	GoodsVo	

5.17.4 GoodsVo Entity Description

5.17.4.1 Entity Name

Entity Name	GoodsVo
Description	

5.17.4.2 Entity Attribute

Attribute	Type	Mandatory	Description	Remarks
orderRelationId	Varchar	Y	Goods Code	It is mandatory
effectiveDate	Long	Y	Effective Time	It is mandatory. Timestamp in GMT0, eg: 1447225157471
expiryDate	Long	Y	Expire Time	It is mandatory. Timestamp in GMT0, eg: 1447225157471

5.17.5 Output Parameter

Attribute	Type	Mandatory	Description	Remarks
streamNo	Char(25)	Y	Stream No	Code (5 bits) + yyyyymmddhhmmss (a 14-bit time sequence) +6-bit sequence eg:UKAPP201511131 61019000001
resultCode	Varchar	Y	Result Code	00000000
resultDesc	Varchar	Y	Result Description	
data	T	Y	data	Data is in JSON format

6 Process for APIs Application

Partner need apply API via email or document currently. After check partner certification and contract, platform administrator will open APIs for partner as soon as possible.

Confidential & uCloudlink

7 Test Case for Reference

Before test, please read section 2.4 business scenarios to understand the business process and read 3.2 technical proposal to understand how to invoke the interface.

7.1 Test Step

Primary Part (Default provide)		
NO.	Interface name	Interface ID
1	Business customer login in	GrpUserLogin
2	Business customer login out	GrpUserLogout
3	Batch create sub-user of business customer	BatchCreateGroupChild
4	Query sub-user list of business customer	QuerySubUserListInfo
5	Query available offer list for sale	QueryGrpOfferList
6	Create order for sub-user	GrpCreateOrder
7	Query offer list of sub-user	QueryUserOfferList
Secondary Part (Open only after application)		
NO.	Interface name	Interface ID
1	Stop and start sub-user (batch)	StopSubUser
2	Delete and resume sub-user (batch)	DeleteSubUser
3	Reset sub-user password	ResetPwdForSubUser
4	Bind device with sub-user	BindDevice
5	Unbind device with sub-user (batch)	BatchUnBindDevice
6	Query device binding Information	QueryBindingRelationInfo
7	Lock and unlock device (batch)	LockDevice
8	Top up for sub-user	TopUpForSubUser
9	Query top up list of sub-user	QueryTopUpListInfo
10	Query account detail of sub-user	QueryAccountDetailListInfo
11	Batch query sub-user list of business customer	QuerySubUserListByBatch
12	Operate offer for sub-user (Old)	SyncOrderRelationFromUP
13	Query offer list of sub-user by orderId	QueryOrderRelationByOrderId
14	Unsubscribe order for sub-user	CancelOrder
15	Operate OrderRelation for sub-user (New)	OperateOrderRelation
16	Disable OrderRelation for sub-user (New)	DisableOrderRelation
17	Update OrderRelation for sub-user (New)	UpdateOrderRelation

Please following below test steps:

7.2 Primary Part's Test Scenarios

7.2.1 Business customer login in

7.2.1.1 Call Entity

<https://saas.ucloudlink.com/bss/grp/noauth/GrpUserLogin>

7.2.1.2 Input Parameter

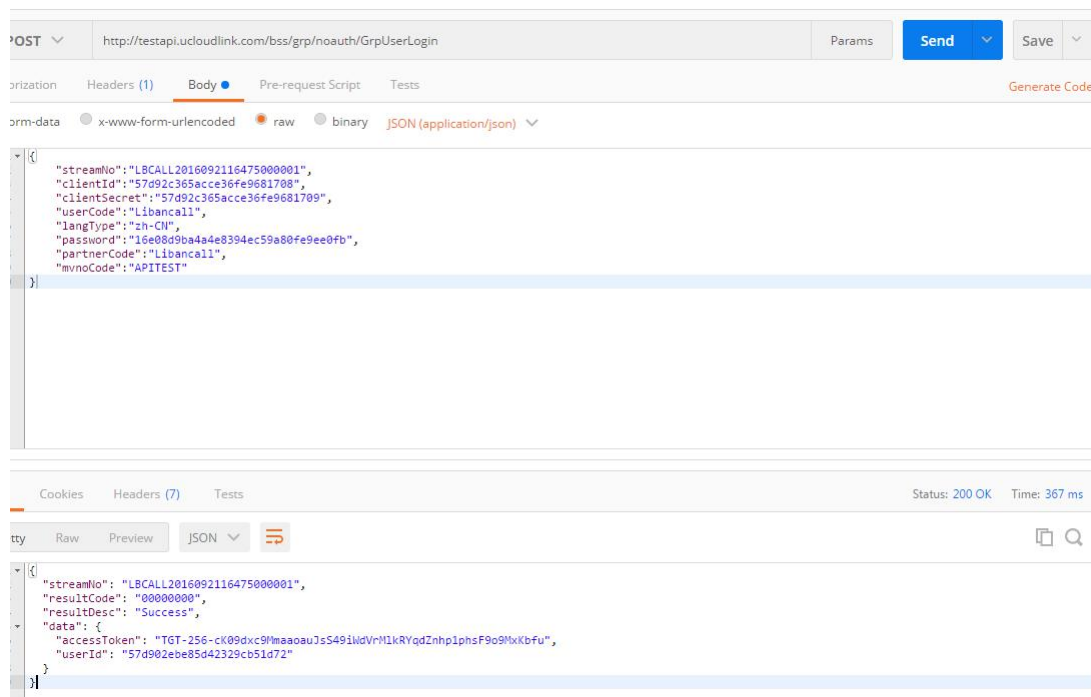
```
{
  "streamNo": "PTNERX2016092116475000001",
  "clientId": "57d92c365acce36fe9681708",
  "clientSecret": "xxxxxxxxxxxxxxxxxxxxxx", ---- please use real client secret
  "userCode": "Partnerx",
  "langType": "zh-CN",
  "password": "xxxxxxxxxxxxxxxxxxxxxx", ---- please use real password and encrypt in MD5 once
  "partnerCode": "Partnerx",
  "mvnoCode": "APITEST"
}
```

7.2.1.3 Output Parameter

```
{
  "streamNo": "PTNERX2016092116475000001",
  "resultCode": "00000000",
  "resultDesc": "Success",
  "data": {
    "accessToken": "TGT-256-cK09dxc9MmaaoauJsS49iWdVrM1kRYqdZnhp1phsF9o9MxKbfu",
    "userId": "57d902ebe85d42329cb51d72"
  }
}
```

Note: accessToken and userId could be used in flowing interfaces.
userId is loginCustomerId in flowing interfaces input parameters.

7.2.1.4 Test Example



7.2.2 Business customer login out

7.2.2.1 Call Entity

https://saas.ucloudlink.com/bss/grp/user/GrpUserLogout?access_token=TGT-266-NHYr7IZutcrGEdJ0qEfbDYWgZzoEgR1MWTMbZlfsQLFuUjIZIG

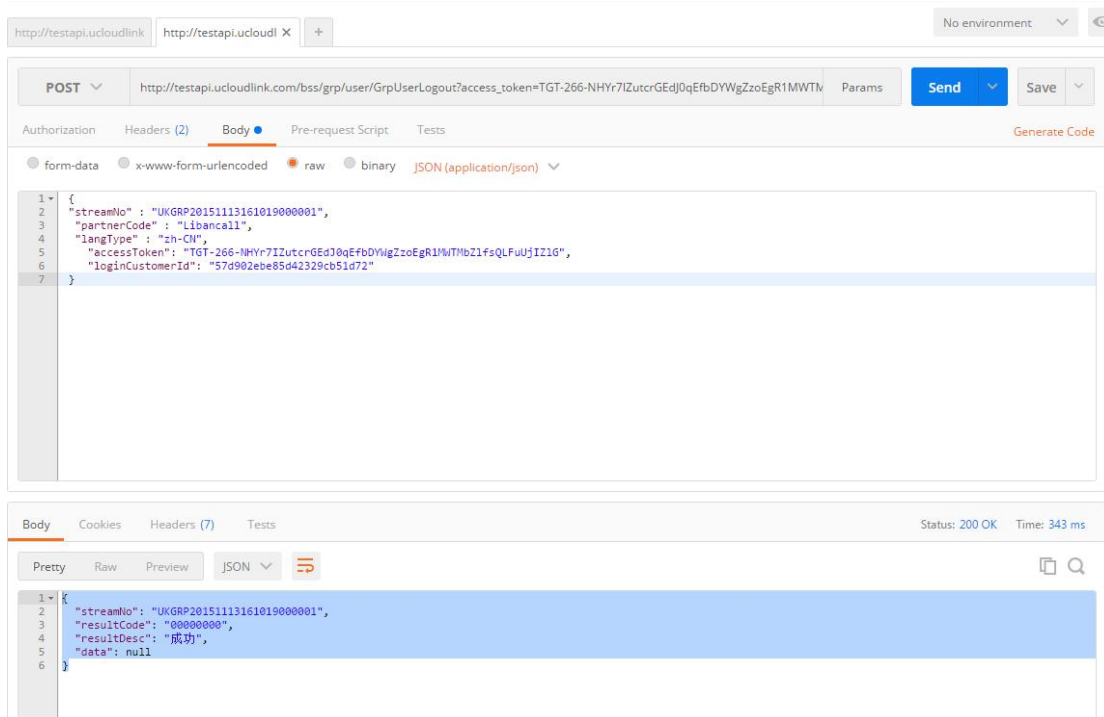
7.2.2.2 Input Parameter

```
{
  "streamNo": "UKGRP20151113161019000001",
  "partnerCode": "Partnerx",
  "langType": "zh-CN",
  "accessToken": "TGT-266-NHYr7IZutcrGEdJ0qEfbDYWgZzoEgR1MWTMbZlfsQLFuUjIZIG",
  "loginCustomerId": "57d902ebe85d42329cb51d72"
}
```

7.2.2.3 Output Parameter

```
{
  "streamNo": "UKGRP20151113161019000001",
  "resultCode": "00000000",
  "resultDesc": "成功",
  "data": null
}
```

7.2.2.4 Test Example



7.2.3 Batch create sub-user of business customer

7.2.3.1 Call Entity

https://saas.ucloudlink.com/bss/grp/user/BatchCreateGroupChild?access_token=TGT-256-cK09dxc9MmaaoauJsS49iWdVrMlkRYqdZnhp1phsF9o9MxKbfu

7.2.3.2 Input Parameter

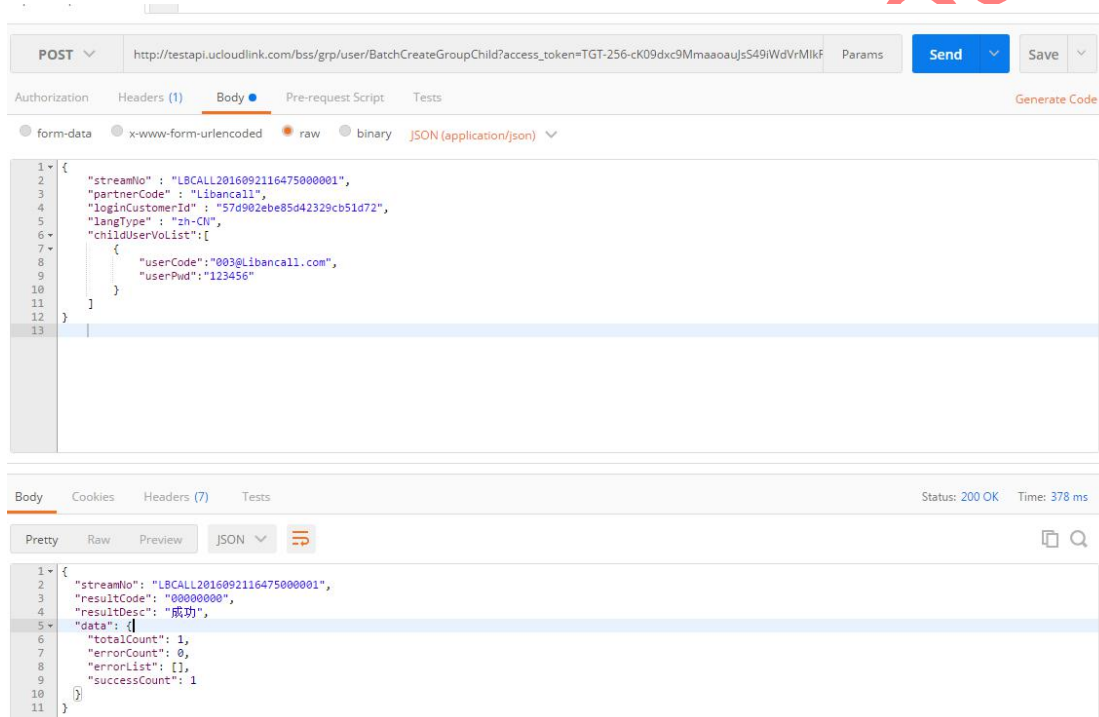
```
{
  "streamNo": "PTNERX2016092116475000001",
  "partnerCode": "Partnerx",
  "loginCustomerId": "57d902ebe85d42329cb51d72",
  "langType": "zh-CN",
  "childUserVoList": [
    {
      "userCode": "003@Partnerx.com",
      "userPwd": "123456",
      "registerType": "EMAIL"
    }
  ]
}
```

Note: If input parameters: imei and tmlPwd, the interface will create sub user and active device with sub user at same time.

7.2.3.3 Output Parameter

```
{
  "streamNo": "PTNERX2016092116475000001",
  "resultCode": "00000000",
  "resultDesc": "成功",
  "data": {
    "totalCount": 1,
    "errorCount": 0,
    "errorList": [],
    "successCount": 1
  }
}
```

7.2.3.4 Test Example



7.2.4 Query sub-user list of business customer

7.2.4.1 Call Entity

https://saas.ucloudlink.com/bss/grp/user/QuerySubUserListInfo?access_token=TGT-256-cK09dxc9MmaaouJsS49iWdVrMikRYqdZnhp1phsF9o9MxKbfu

7.2.4.2 Input Parameter

```
{
```

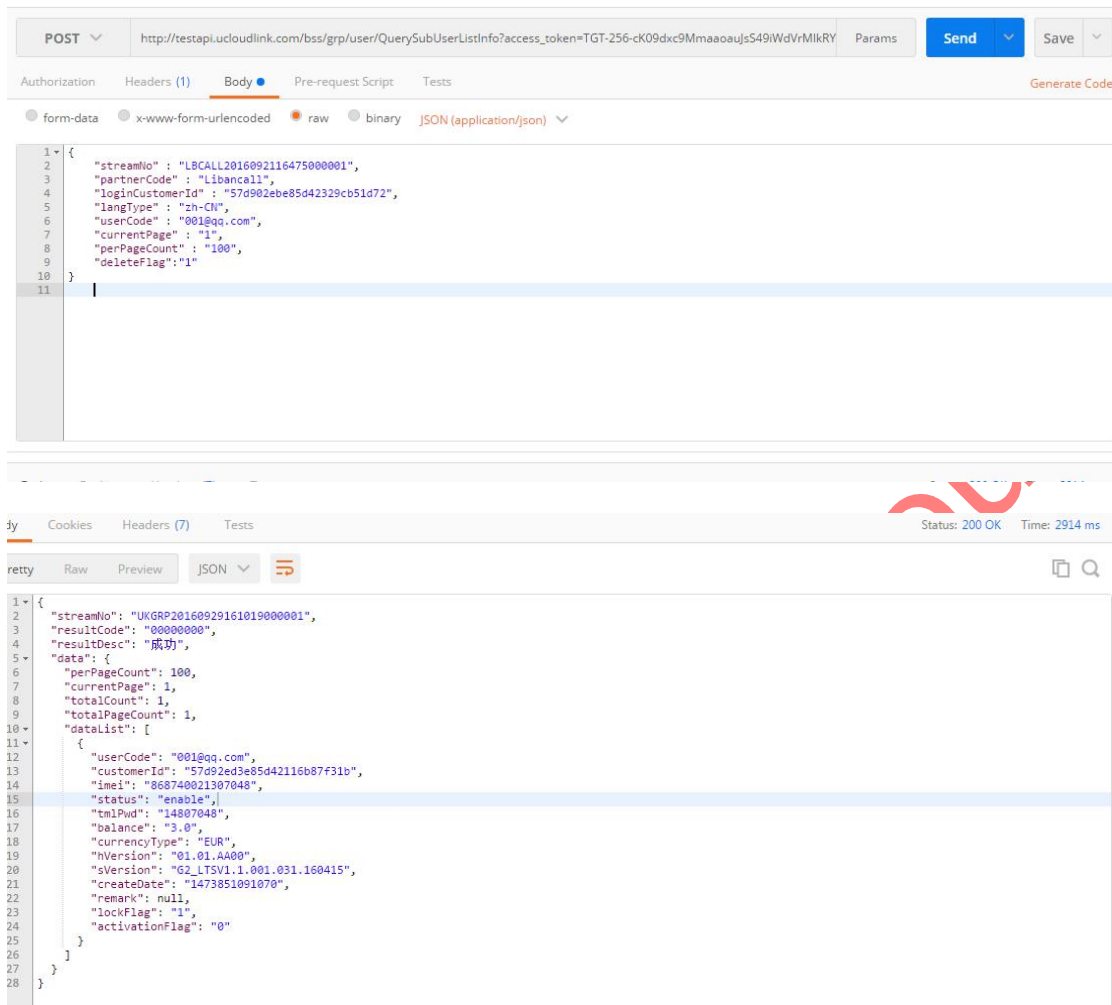
```
"streamNo": "PTNERX2016092116475000001",  
"partnerCode": "Partnerx",  
"loginCustomerId": "57d902ebe85d42329cb51d72",  
"langType": "zh-CN",  
"userCode": "001@qq.com",  
"currentPage": "1",  
"perPageCount": "100",  
"deleteFlag": "1"  
}
```

Note: usercode could be null, then the interface will return all sub users in list.

7.2.4.3 Output Parameter

```
{  
  "streamNo": "UKGRP20160929161019000001",  
  "resultCode": "00000000",  
  "resultDesc": "成功",  
  "data": {  
    "perPageCount": 100,  
    "currentPage": 1,  
    "totalCount": 1,  
    "totalPageCount": 1,  
    "dataList": [  
      {  
        "userCode": "001@qq.com",  
        "customerId": "57d92ed3e85d42116b87f31b",  
        "imei": "868740021307048",  
        "status": "enable",  
        "tmIPwd": "14807048",  
        "balance": "3.0",  
        "currencyType": "EUR",  
        "hVersion": "01.01.AA00",  
        "sVersion": "G2_LTSV1.1.001.031.160415",  
        "createDate": "1473851091070",  
        "remark": null,  
        "lockFlag": "1",  
        "activationFlag": "0"  
      }  
    ]  
  }  
}
```

7.2.4.4 Test Example



The screenshot displays a REST client interface with a POST request and its corresponding JSON response.

Request:

- Method: POST
- URL: `http://testapi.ucloudlink.com/bss/grp/user/QuerySubUserListInfo?access_token=TGT-256-cK09dxc9MmaaoauJs49iWdVrMikRY`
- Body (raw):


```
{
  "streamNo": "LBCALL2016092116475000001",
  "partnerCode": "Libancall1",
  "loginCustomerId": "57d902ebe85d42329cb51d72",
  "langType": "zh-CN",
  "userCode": "001@qq.com",
  "currentPage": "1",
  "perPageCount": "100",
  "deleteFlag": "1"
}
```

Response:

- Status: 200 OK
- Time: 2914 ms
- Body (raw):


```
{
  "streamNo": "UKGRP20160929161019000001",
  "resultCode": "00000000",
  "resultDesc": "成功",
  "data": {
    "perPageCount": 100,
    "currentPage": 1,
    "totalCount": 1,
    "totalPageCount": 1,
    "dataList": [
      {
        "userCode": "001@qq.com",
        "customerId": "57d92ed3e85d42116b87f31b",
        "imei": "868740021307048",
        "status": "enable",
        "tm1Pw": "14807048",
        "balance": "3.0",
        "currencyType": "EUR",
        "hVersion": "01.01.AA00",
        "sVersion": "G2_LTSV1.1.001.031.160415",
        "createDate": "1473851091070",
        "remark": null,
        "lockFlag": "1",
        "activationFlag": "0"
      }
    ]
  }
}
```

7.2.5 Query available offer list for sale

7.2.5.1 Call Entity

`https://saas.ucloudlink.com/bss/grp/goods/QueryGrpOfferList?access_token=TGT-256-cK09dxc9MmaaoauJs49iWdVrMikRYqdZnhp1phsF9o9MxKbfu`

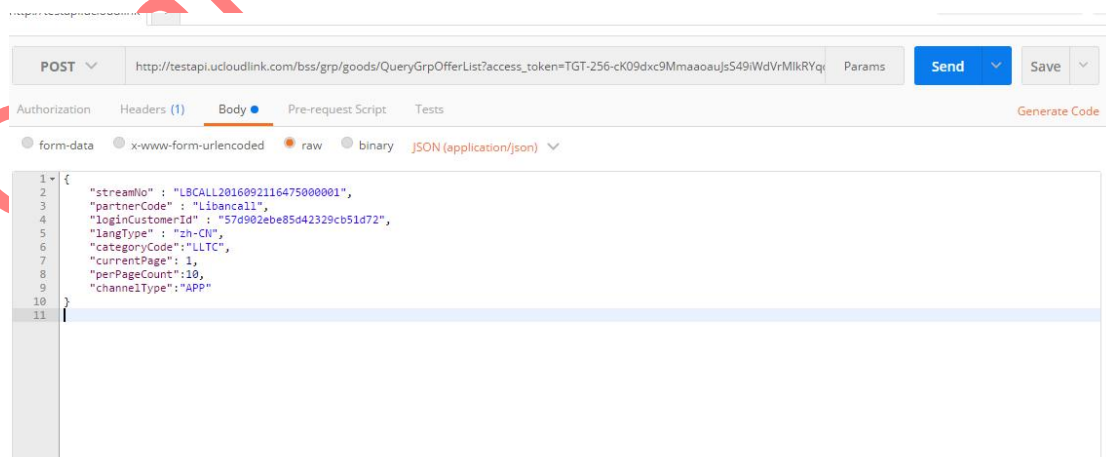
7.2.5.2 Input Parameter

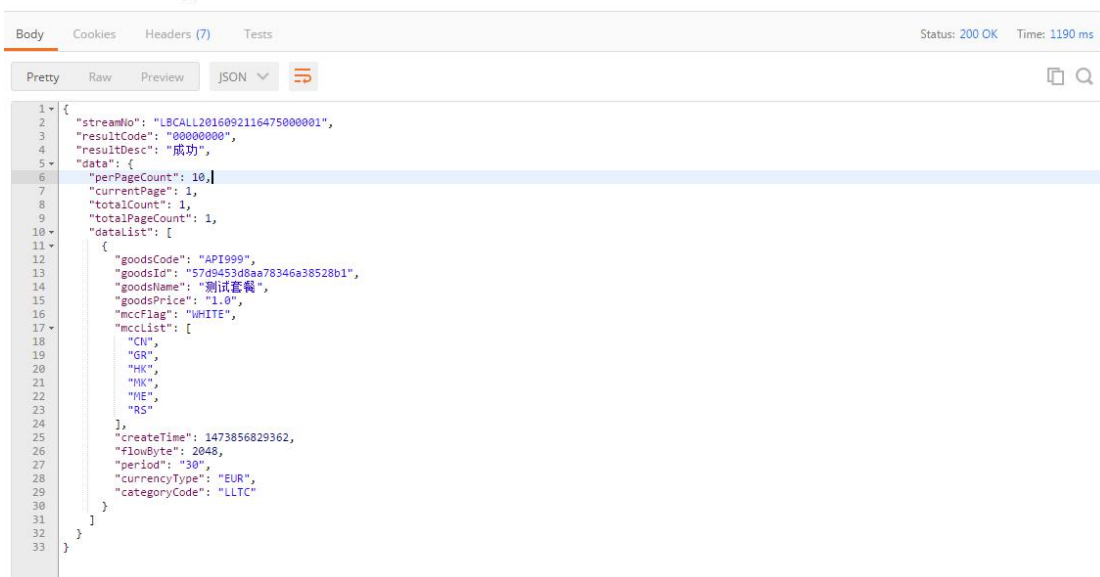
```
{
  "streamNo": "PTNERX2016092116475000001",
  "partnerCode": "Partnerx",
  "loginCustomerId": "57d902ebe85d42329cb51d72",
  "langType": "zh-CN",
  "categoryCode": "LLTC",
  "currentPage": 1,
  "perPageCount": 10,
  "channelType": "GRP"
}
```

7.2.5.3 Output Parameter

```
{
  "streamNo": "PTNERX2016092116475000001",
  "resultCode": "00000000",
  "resultDesc": "成功",
  "data": {
    "perPageCount": 10,
    "currentPage": 1,
    "totalCount": 1,
    "totalPageCount": 1,
    "dataList": [
      {
        "goodsCode": "API999",
        "goodsId": "57d9453d8aa78346a38528b1",
        "goodsName": "测试套餐",
        "goodsPrice": "1.0",
        "mccFlag": "WHITE",
        "mccList": [
          "CN",
          "GR",
          "HK",
          "MK",
          "ME",
          "RS"
        ],
        "createTime": 1473856829362,
        "flowByte": 2048,
        "period": "30",
        "currencyType": "EUR",
        "categoryCode": "LLTC"
      }
    ]
  }
}
```

7.2.5.4 Test Example





7.2.6 Create order for sub-user

7.2.6.1 Call Entity

https://saas.ucloudlink.com/bss/grp/order/GrpCreateOrder?access_token=TGT-256-cK09dxc9MmaaoauJsS49iWdVrMikRYqdZnhp1phsF9o9MxKbfu

7.2.6.2 Input Parameter

```

{
  "streamNo": "PTNERX2016092116475000001",
  "partnerCode": "Partnerx",
  "loginCustomerId": "57d902ebe85d42329cb51d72",
  "userCode": "001@qq.com",
  "channelType": "GRP",
  "orderType": "BUYPKG",
  "payMethod": "ACCOUNT_AMOUNT",
  "orderMark": "test",
  "currencyType": "EUR",
  "goodsList": [
    {
      "goodsId": "57d9453d8aa78346a38528b1",
      "quantity": 1
    }
  ]
}
  
```

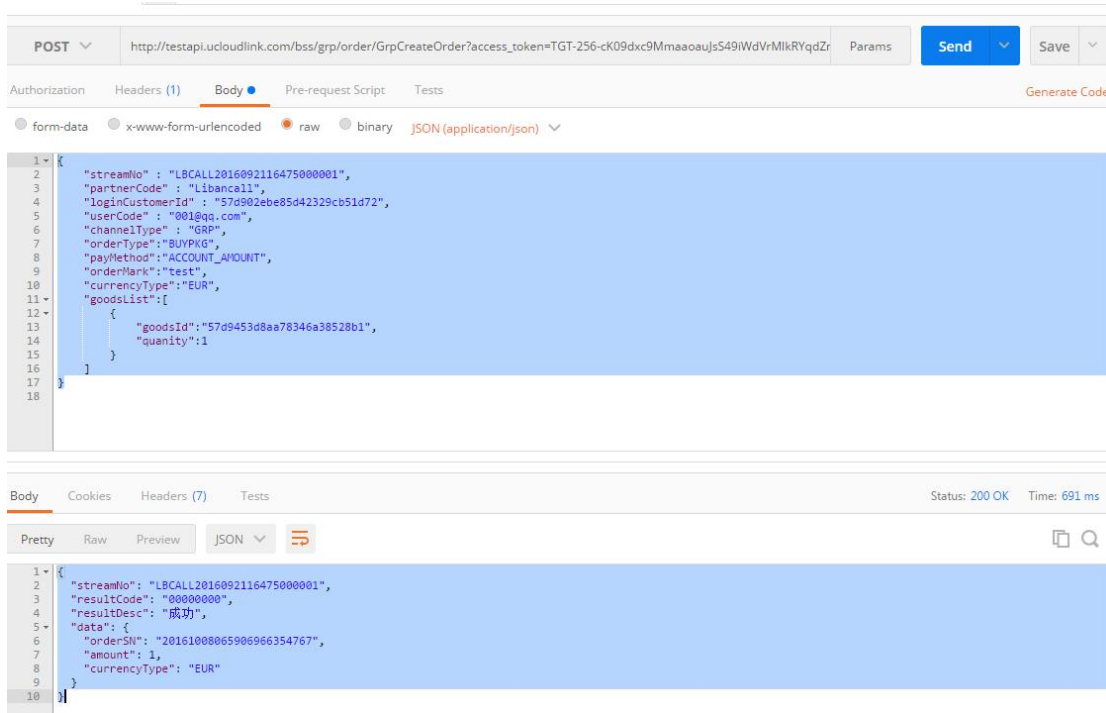
7.2.6.3 Output Parameter

```

{
  "streamNo": "PTNERX2016092116475000001",
  "resultCode": "00000000",
  "resultDesc": "成功",
  "data": {
    "orderSN": "20161008065906966354767",
    "amount": 1,
  }
}
  
```

```
"currencyType": "EUR"
}
```

7.2.6.4 Test Example



The screenshot shows a REST client interface with a POST request to `http://testapi.ucloudlink.com/bss/grp/order/GrpCreateOrder?access_token=TGT-256-cK09dxc9Mmaaoajs549IWdVrMlkRYqdZr`. The request body is a JSON object with the following fields:

```
{
  "streamNo": "LBCALL2016092116475000001",
  "partnerCode": "Libancall",
  "loginCustomerId": "57d902ebe85d42329cb51d72",
  "userCode": "001@qq.com",
  "channelType": "GRP",
  "orderType": "BUYPG",
  "payMethod": "ACCOUNT_AMOUNT",
  "orderMark": "test",
  "currencyType": "EUR",
  "goodsList": [
    {
      "goodsId": "57d9453d8aa78346a38528b1",
      "quantity": 1
    }
  ]
}
```

The response status is 200 OK with a time of 691 ms. The response body is a JSON object with the following fields:

```
{
  "streamNo": "LBCALL2016092116475000001",
  "resultCode": "00000000",
  "resultDesc": "成功",
  "data": {
    "orderSN": "20161008065906966354767",
    "amount": 1,
    "currencyType": "EUR"
  }
}
```

7.2.7 Query offer list of sub-user

7.2.7.1 Call Entity

https://saas.ucloudlink.com/bss/grp/goods/QueryUserOfferList?access_token=TGT-266-NHYr7IZutrGEdJ0qEfbDYWgZzoEgR1MWTMbZlfsQLFuUjIZIG

7.2.7.2 Input Parameter

```
{
  "streamNo": "GLCAM20151113161019000001",
  "partnerCode": "GTBUORG",
  "loginCustomerId": "57ce694cf5b95a5444869213",
  "langType": "en-US",
  "userCode": "fangpeng@ukelink.com",
  "flag": "0",
  "goodsType": "DISC",
  "perPageCount": 1,
  "currentPage": 1
}
```

7.2.7.3 Output Parameter

```
{
  "streamNo": "GLCAM20151113161019000001",
  "resultCode": "00000000",
  "resultDesc": "The operation is success.",
  "data": {
    "perPageCount": 1,
    "currentPage": 1,
    "totalCount": 3,
    "totalPageCount": 3,
    "dataList": [
      {
        "relationId": "58afedab26cf67649ab0fdee",
        "customerId": "57e4e375f5b95a55154cd5fd",
        "customerName": "fangpeng@ukelink.com",
        "goodsId": "58748df926cf67371060e318",
        "goodsName": "sellsGood_zjm_0800",
        "status": "VALID",
        "createTime": 1487924651631,
        "mccFlag": "WHITE",
        "mccList": [
          "ALL",
          "HK",
          "CN"
        ],
        "iso2List": null,
        "effectiveTime": 1488297600632,
        "expiryTime": 2434982399632,
        "flowByte": 1024,
        "surplusFlowbyte": null,
        "period": "30",
        "periodUnit": "MONTH",
        "goodsType": "DISC",
        "categoryCode": "CXTC",
        "topFlag": null,
        "accumulatedFlowList": null,
        "attrMap": {
          "pkDesc": "英文1、套餐说明<br>2、套餐说明2<br>3、套餐说明3",
          "effType": "NEXT_MONTH",
          "title": "sellsGood_zjm_0800",
          "infiniFlag": "false",
          "flowSize": "1073741824",
          "timeZone": "GMT+8",
          "period": "30",
          "periodUnit": "MONTH"
        },
        "orderId": null
      }
    ]
  }
}
```

7.3 Secondary Part's Test Scenarios

7.3.1 Stop and start sub-user (batch)

7.3.1.1 Call Entity

https://saas.ucloudlink.com/bss/grp/user/StopSubUser?access_token=TGT-256-cK09dxc9MmaaoauJsS49iWdVrMikRYqdZnhp1phsF9o9MxKbfu

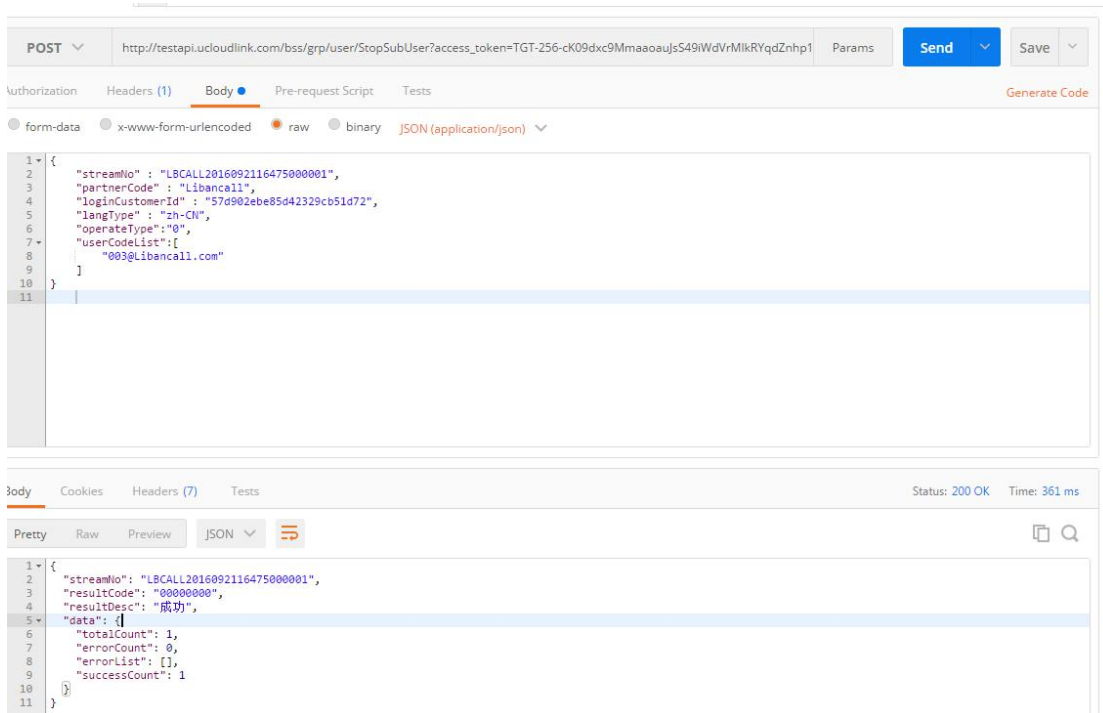
7.3.1.2 Input Parameter

```
{
  "streamNo": "PTNERX2016092116475000001",
  "partnerCode": "Partnerx",
  "loginCustomerId": "57d902ebe85d42329cb51d72",
  "langType": "zh-CN",
  "operateType": "0",
  "userCodeList": [
    "003@Partnerx.com"
  ]
}
```

7.3.1.3 Output Parameter

```
{
  "streamNo": "PTNERX2016092116475000001",
  "resultCode": "00000000",
  "resultDesc": "成功",
  "data": {
    "totalCount": 1,
    "errorCount": 0,
    "errorList": [],
    "successCount": 1
  }
}
```

7.3.1.4 Test Example



7.3.2 Delete and resume sub-user (batch)

7.3.2.1 Call Entity

https://saas.ucloudlink.com/bss/grp/user/DeleteSubUser?access_token=TGT-256-cK09dxc9MmaaoauJs49iWdVrMikRYqdZnhp1phsF9o9MxKbfu

7.3.2.2 Input Parameter

```

{
  "streamNo": "PTNERX2016092116475000001",
  "partnerCode": "Partnerx",
  "loginCustomerId": "57d902ebe85d42329cb51d72",
  "langType": "zh-CN",
  "operateType": "0",
  "userCodeList": [
    "003@Partnerx.com"
  ]
}

```

7.3.2.3 Output Parameter

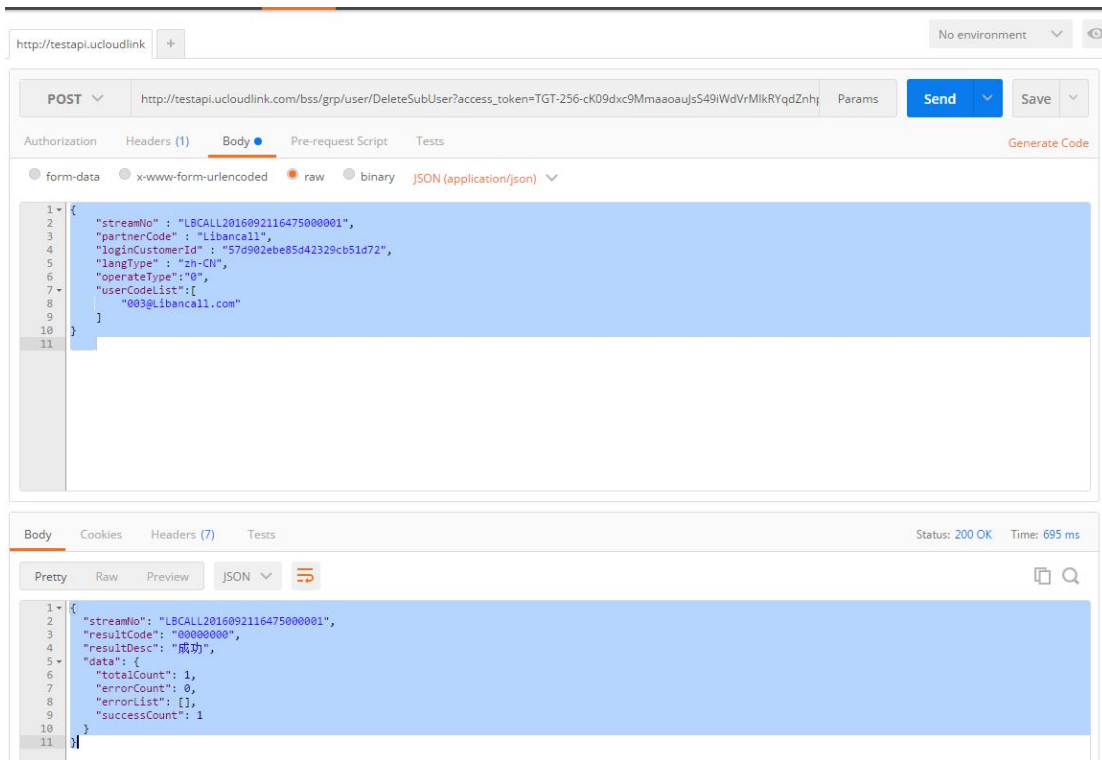
```

{
  "streamNo": "PTNERX2016092116475000001",
  "resultCode": "00000000",
  "resultDesc": "成功",
  "data": {

```

```
"totalCount": 1,
"errorCount": 0,
"errorList": [],
"successCount": 1
}
```

7.3.2.4 Test Example



7.3.3 Reset sub-user password

7.3.3.1 Call Entity

https://saas.ucloudlink.com/bss/grp/user/ResetPwdForSubUser?access_token=TGT-256-cK09dxc9Mmaaoajs49iWdVrMikRYqdZnhp1phsF9o9MxKbfu

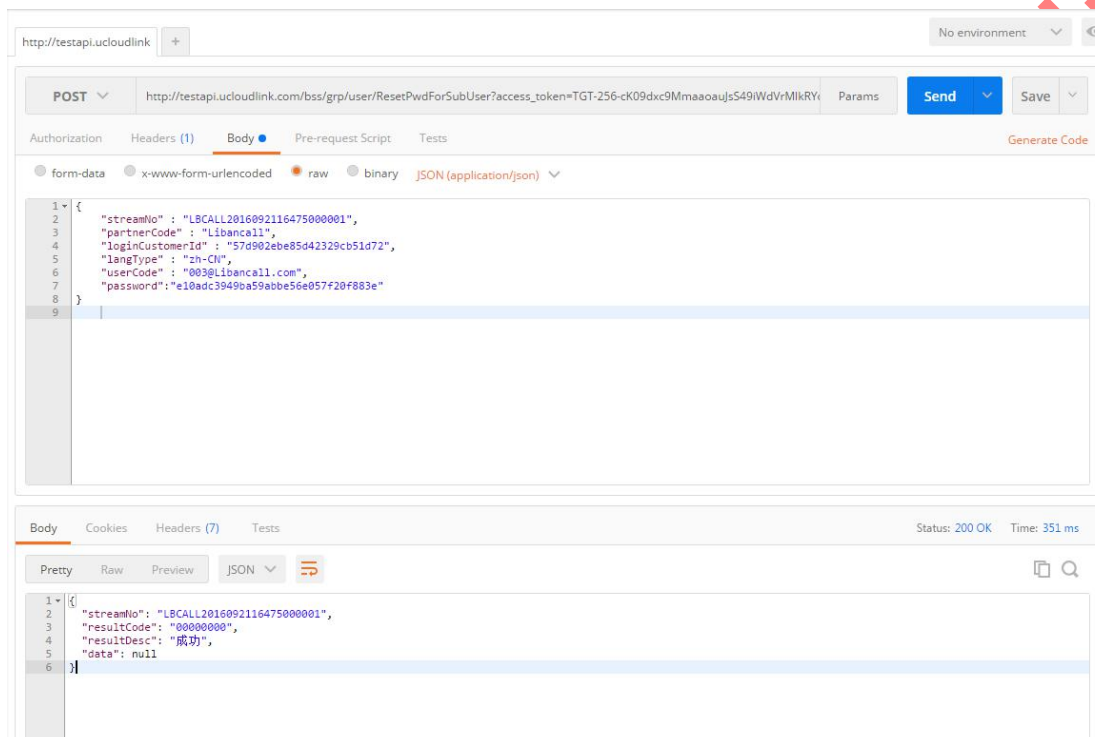
7.3.3.2 Input Parameter

```
{
  "streamNo": "PTNERX2016092116475000001",
  "partnerCode": "Partnerx",
  "loginCustomerId": "57d902ebe85d42329cb51d72",
  "langType": "zh-CN",
  "userCode": "003@Partnerx.com",
  "password": "e10adc3949ba59abbe56e057f20f883e"
}
```

7.3.3.3 Output Parameter

```
{
  "streamNo": "PTNERX2016092116475000001",
  "resultCode": "00000000",
  "resultDesc": "成功",
  "data": null
}
```

7.3.3.4 Test Example



7.3.4 Bind device with sub-user

7.3.4.1 Call Entity

https://saas.ucloudlink.com/bss/grp/tml/BindDevice?access_token=TGT-256-ck09dxc9MmaaouJsS49iWdVrMikRYqdZnhp1phsF9o9MxKbfu

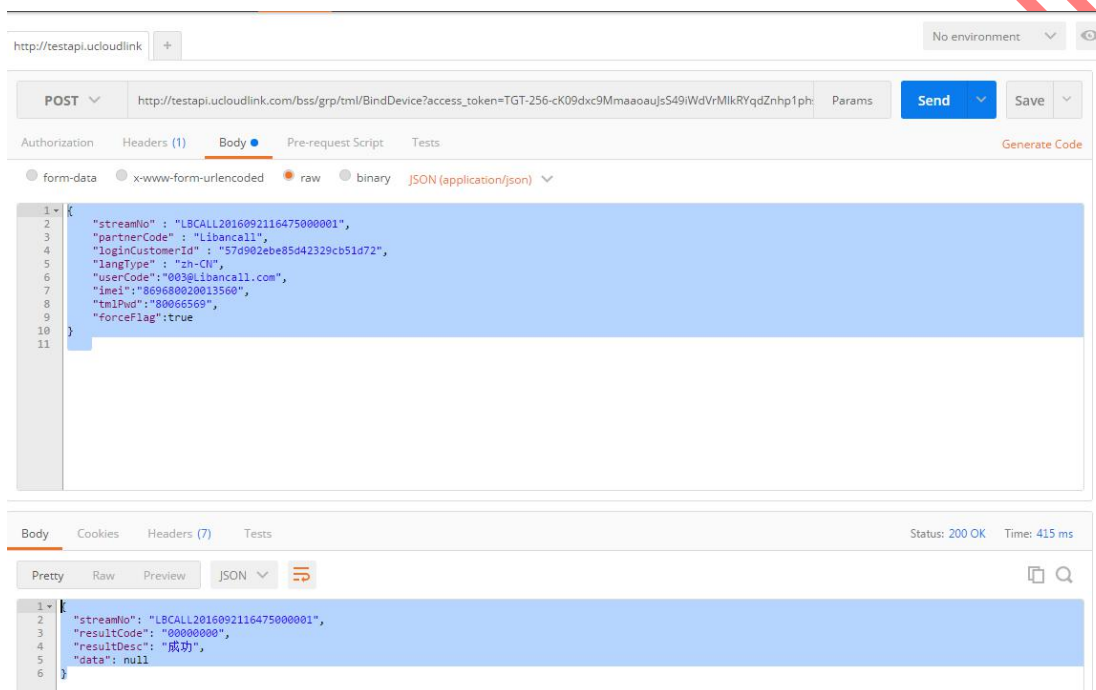
7.3.4.2 Input Parameter

```
{
  "streamNo" : "PTNERX2016092116475000001",
  "partnerCode" : "Partnerx",
  "loginCustomerId" : "57d902ebe85d42329cb51d72",
  "langType" : "zh-CN",
  "userCode": "003@Partnerx.com",
  "imei": "869680020013560",
  "tmlPwd": "80066569",
  "forceFlag": true
}
```

7.3.4.3 Output Parameter

```
{
  "streamNo": "PTNERX2016092116475000001",
  "resultCode": "00000000",
  "resultDesc": "成功",
  "data": null
}
```

7.3.4.4 Test Example



7.3.5 Unbind device with sub-user (batch)

7.3.5.1 Call Entity

https://saas.ucloudlink.com/bss/grp/tml/BatchUnBindDevice?access_token=TGT-256-cK09dxc9MmaaouJsS49iWdVrMikRYqdZnhp1phsF9o9MxKbfu

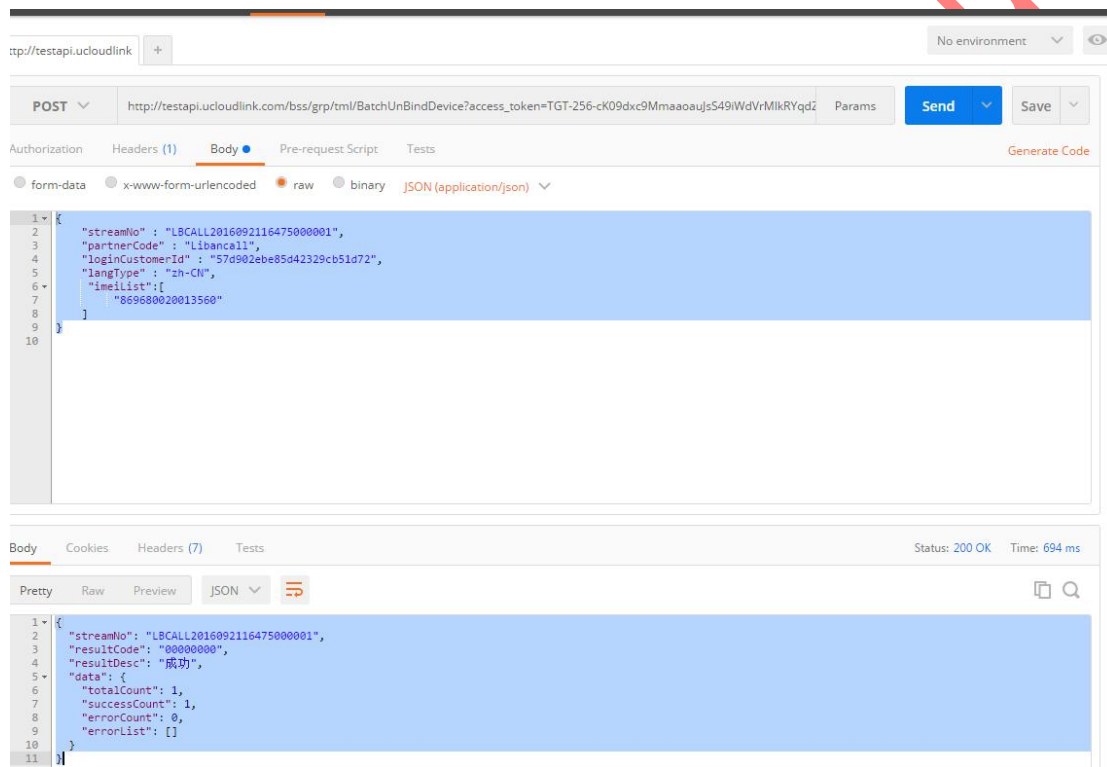
7.3.5.2 Input Parameter

```
{
  "streamNo": "PTNERX2016092116475000001",
  "partnerCode": "Partnerx",
  "loginCustomerId": "57d902ebe85d42329cb51d72",
  "langType": "zh-CN",
  "imeiList": [
    "869680020013560"
  ]
}
```

7.3.5.3 Output Parameter

```
{
  "streamNo": "PTNERX2016092116475000001",
  "resultCode": "00000000",
  "resultDesc": "成功",
  "data": {
    "totalCount": 1,
    "successCount": 1,
    "errorCount": 0,
    "errorList": []
  }
}
```

7.3.5.4 Test Example



7.3.6 Query device binding Information

7.3.6.1 Call Entity

https://saas.ucloudlink.com/bss/grp/tml/QueryBindingRelationInfo?access_token=TGT-256-cK09dxc9MmaaoauJs49iWdVrMikRYqdzZnhp1phsF9o9MxKbfu

7.3.6.2 Input Parameter

```
{
  "streamNo": "PTNERX2016092116475000001",
  "partnerCode": "Partnerx",
  "loginCustomerId": "57d902ebe85d42329cb51d72",
}
```

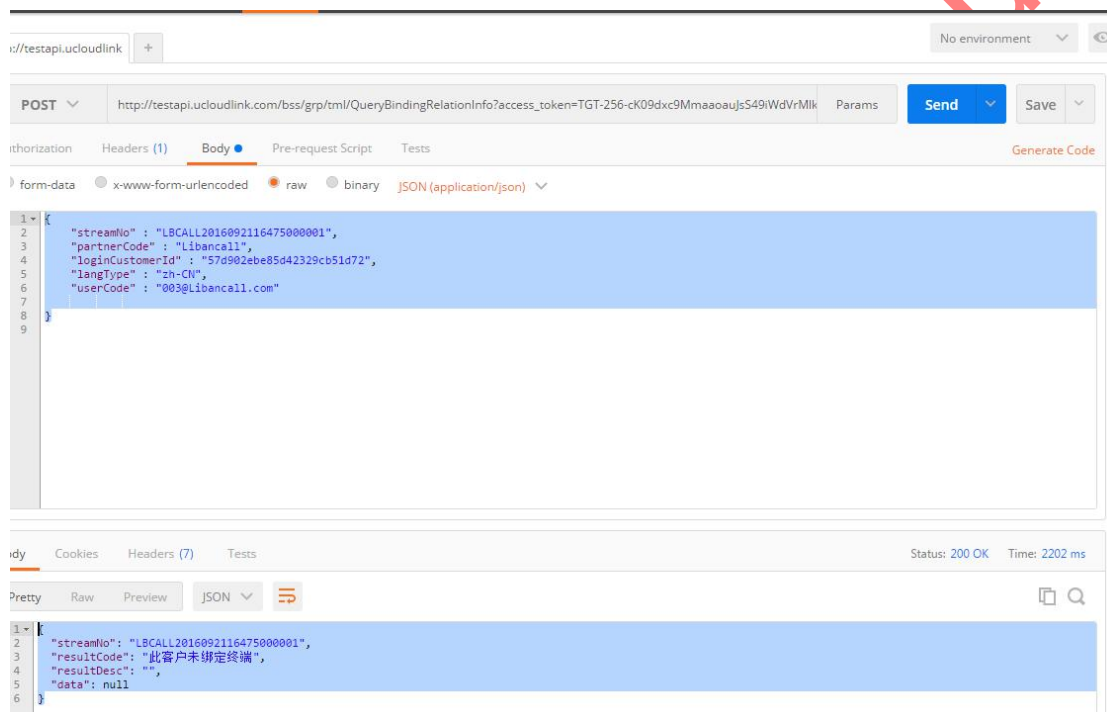
```
"langType" : "zh-CN",
"userCode" : "003@Partnerx.com"

}
```

7.3.6.3 Output Parameter

```
{
  "streamNo": "PTNERX2016092116475000001",
  "resultCode": "此客户未绑定终端",
  "resultDesc": "",
  "data": null
}
```

7.3.6.4 Test Example



7.3.7 Lock and unlock device (batch)

7.3.7.1 Call Entity

https://saas.ucloudlink.com/bss/grp/tml/LockDevice?access_token=TGT-256-cK09dxc9MmaaoauJsS49iWdVrMikRYqdZnhp1phsF9o9MxKbfu

7.3.7.2 Input Parameter

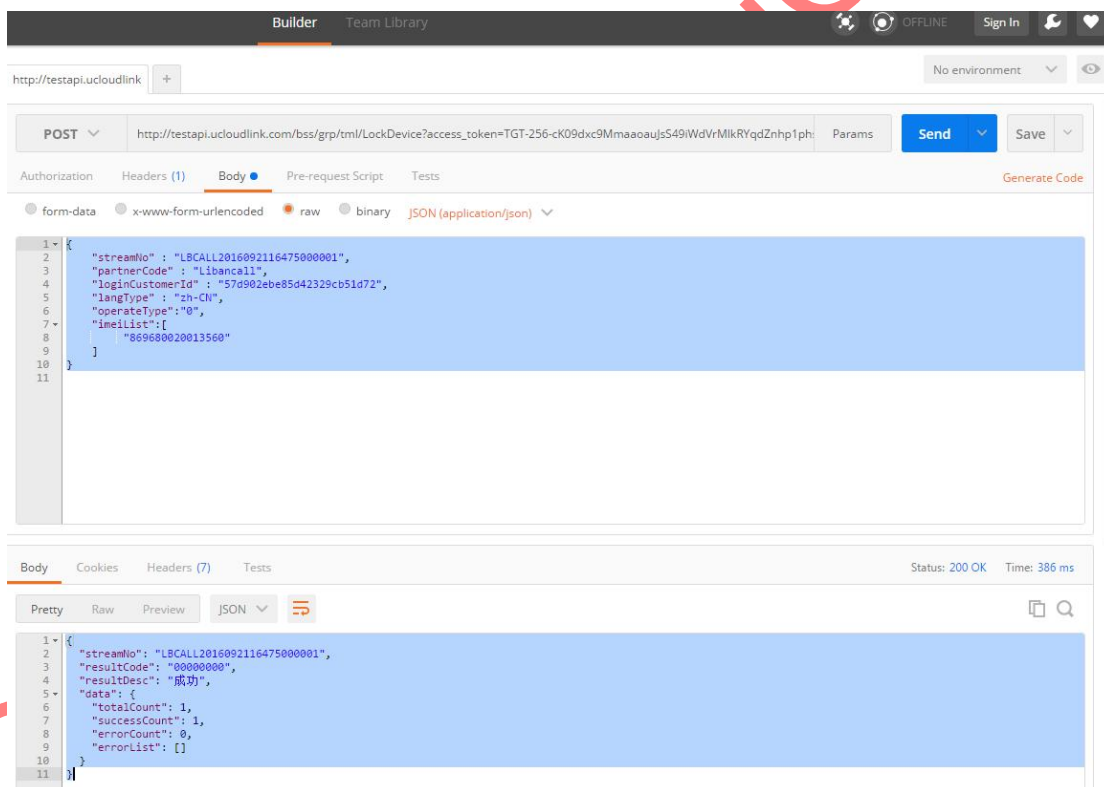
```
{
  "streamNo": "PTNERX2016092116475000001",
  "partnerCode": "Partnerx",
  "loginCustomerId": "57d902ebe85d42329cb51d72",
  "langType": "zh-CN",
}
```

```
"operateType": "0",
"imeiList": [
  "869680020013560"
]
```

7.3.7.3 Output Parameter

```
{
  "streamNo": "PTNERX2016092116475000001",
  "resultCode": "00000000",
  "resultDesc": "成功",
  "data": {
    "totalCount": 1,
    "successCount": 1,
    "errorCount": 0,
    "errorList": []
  }
}
```

7.3.7.4 Test Example



7.3.8 Top up for sub-user

7.3.8.1 Call Entity

https://saas.ucloudlink.com/bss/grp/topup/TopUpForSubUser?access_token=TGT-256-cK09dxc9MmaaouJs49iWdVrMikRYqdZnhp1phsF9o9MxKbfu

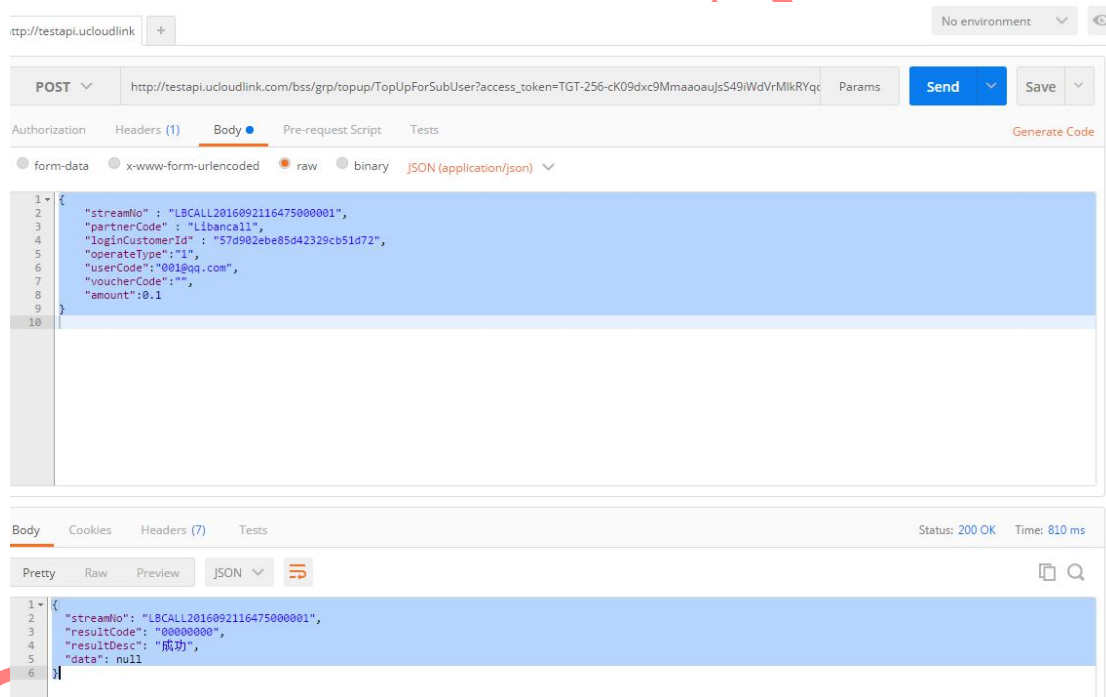
7.3.8.2 Input Parameter

```
{
  "streamNo": "PTNERX2016092116475000001",
  "partnerCode": "Partnerx",
  "loginCustomerId": "57d902ebe85d42329cb51d72",
  "operateType": "1",
  "userCode": "001@qq.com",
  "amount": 0.1
}
```

7.3.8.3 Output Parameter

```
{
  "streamNo": "PTNERX2016092116475000001",
  "resultCode": "00000000",
  "resultDesc": "成功",
  "data": null
}
```

7.3.8.4 Test Example



7.3.9 Query top up list of sub-user

7.3.9.1 Call Entity

https://saas.ucloudlink.com/bss/grp/topup/QueryTopUpListInfo?access_token=TGT-256-cK09dxc9MmaaoauJs49iWdVrMikRYqdZnhp1phsF9o9MxKbfu

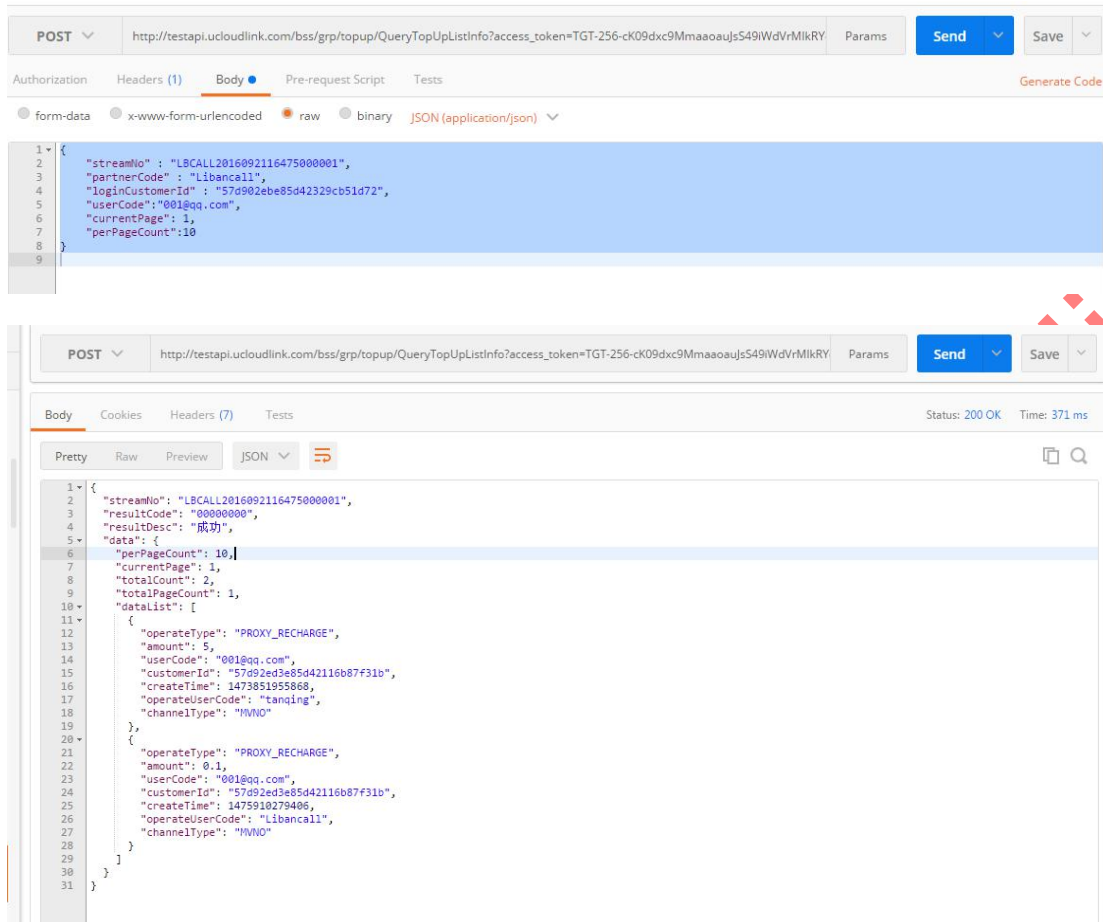
7.3.9.2 Input Parameter

```
{
  "streamNo": "PTNERX2016092116475000001",
  "partnerCode": "Partnerx",
  "loginCustomerId": "57d902ebe85d42329cb51d72",
  "userCode": "001@qq.com",
  "currentPage": 1,
  "perPageCount": 10
}
```

7.3.9.3 Output Parameter

```
{
  "streamNo": "PTNERX2016092116475000001",
  "resultCode": "00000000",
  "resultDesc": "成功",
  "data": {
    "perPageCount": 10,
    "currentPage": 1,
    "totalCount": 2,
    "totalPageCount": 1,
    "dataList": [
      {
        "operateType": "PROXY_RECHARGE",
        "amount": 5,
        "userCode": "001@qq.com",
        "customerId": "57d92ed3e85d42116b87f31b",
        "createTime": 1473851955868,
        "operateUserCode": "tanqing",
        "channelType": "MVNO"
      },
      {
        "operateType": "PROXY_RECHARGE",
        "amount": 0.1,
        "userCode": "001@qq.com",
        "customerId": "57d92ed3e85d42116b87f31b",
        "createTime": 1475910279406,
        "operateUserCode": "Partnerx",
        "channelType": "MVNO"
      }
    ]
  }
}
```

7.3.9.4 Test Example



The first screenshot shows a POST request to `http://testapi.ucloudlink.com/bss/grp/topup/QueryTopUpListInfo?access_token=TGT-256-ck09dxc9MmaaoauJsS49iWdVrMikRY`. The request body is a JSON object:

```
{
  "streamNo": "LBCALL2016092116475000001",
  "partnerCode": "Libancall",
  "loginCustomerId": "57d902ebe85d42329cb51d72",
  "userCode": "001@qq.com",
  "currentPage": 1,
  "perPageCount": 10
}
```

The second screenshot shows the response body, which is a JSON object with status 200 OK and time 371 ms:

```
{
  "resultCode": "00000000",
  "resultDesc": "成功",
  "data": {
    "perPageCount": 10,
    "currentPage": 1,
    "totalCount": 2,
    "totalPageCount": 1,
    "dataList": [
      {
        "operateType": "PROXY_RECHARGE",
        "amount": 5,
        "userCode": "001@qq.com",
        "customerId": "57d92ed3e85d42116b87f31b",
        "createTime": 1473851955868,
        "operateUserCode": "tanqing",
        "channelType": "MVNO"
      },
      {
        "operateType": "PROXY_RECHARGE",
        "amount": 0.1,
        "userCode": "001@qq.com",
        "customerId": "57d92ed3e85d42116b87f31b",
        "createTime": 1475910279406,
        "operateUserCode": "Libancall",
        "channelType": "MVNO"
      }
    ]
  }
}
```

7.3.10 Query account detail of sub-user

7.3.10.1 Call Entity

https://saas.ucloudlink.com/bss/grp/user/QueryAccountDetailListInfo?access_token=TGT-256-ck09dxc9MmaaoauJsS49iWdVrMikRYqdZnhp1phsF9o9MxKbfu

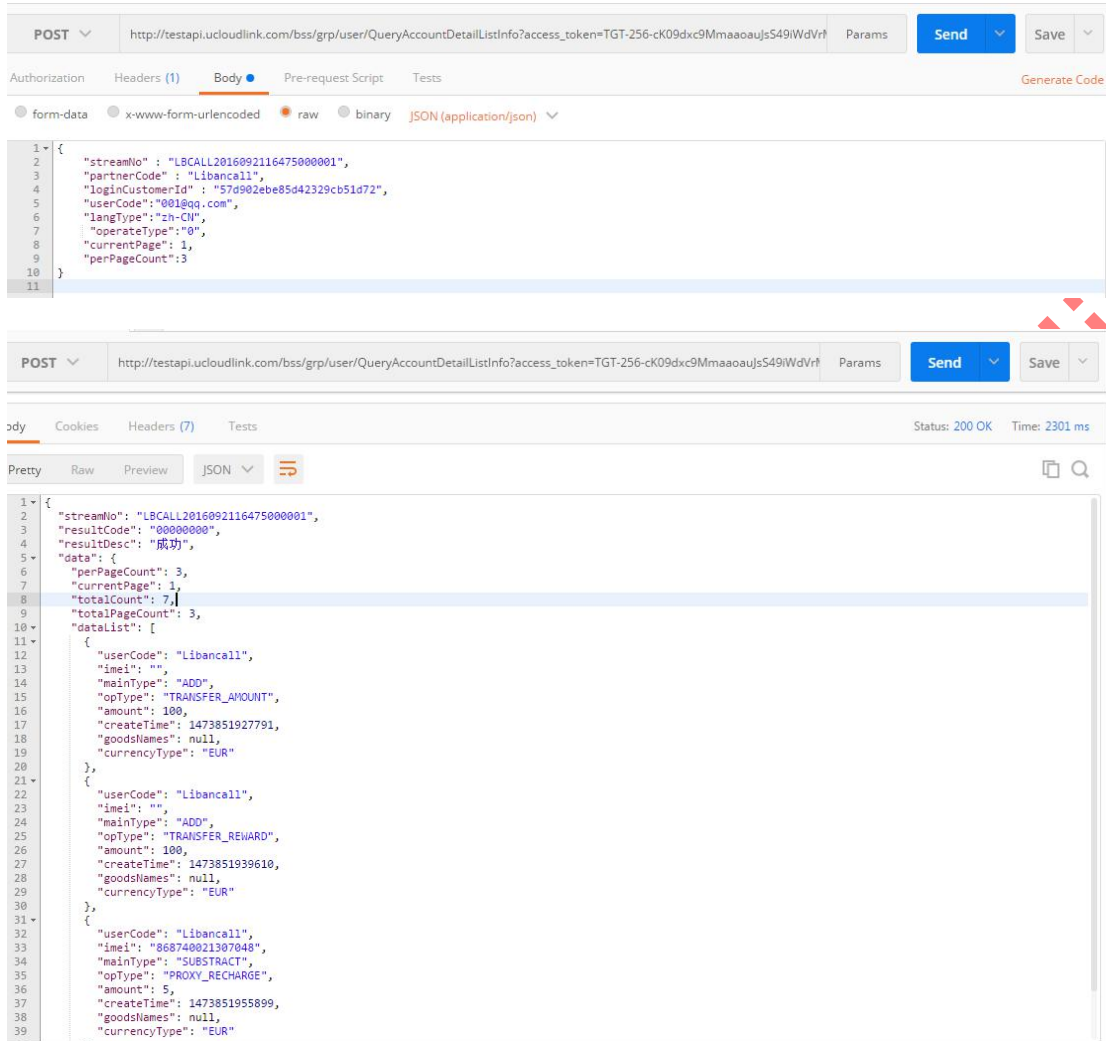
7.3.10.2 Input Parameter

```
{
  "streamNo": "PTNERX2016092116475000001",
  "partnerCode": "Partnerx",
  "loginCustomerId": "57d902ebe85d42329cb51d72",
  "userCode": "001@qq.com",
  "langType": "zh-CN",
  "operateType": "0",
  "currentPage": 1,
  "perPageCount": 3
}
```

7.3.10.3 Output Parameter

```
{
  "streamNo": "PTNERX2016092116475000001",
  "resultCode": "00000000",
  "resultDesc": "成功",
  "data": {
    "perPageCount": 3,
    "currentPage": 1,
    "totalCount": 7,
    "totalPageCount": 3,
    "dataList": [
      {
        "userCode": "Partnerx",
        "imei": "",
        "mainType": "ADD",
        "opType": "TRANSFER_AMOUNT",
        "amount": 100,
        "createTime": 1473851927791,
        "goodsNames": null,
        "currencyType": "EUR"
      },
      {
        "userCode": "Partnerx",
        "imei": "",
        "mainType": "ADD",
        "opType": "TRANSFER_REWARD",
        "amount": 100,
        "createTime": 1473851939610,
        "goodsNames": null,
        "currencyType": "EUR"
      },
      {
        "userCode": "Partnerx",
        "imei": "868740021307048",
        "mainType": "SUBSTRACT",
        "opType": "PROXY_RECHARGE",
        "amount": 5,
        "createTime": 1473851955899,
        "goodsNames": null,
        "currencyType": "EUR"
      }
    ]
  }
}
```

7.3.10.4 Test Example



The screenshot displays a REST client interface with a POST request to `http://testapi.ucloudlink.com/bss/grp/user/QueryAccountDetailListInfo?access_token=TGT-256-cK09dxc9Mmaaoajs549iWdVr`. The request body is a JSON object with the following fields:

```

{
  "streamNo": "LBCALL2016092116475000001",
  "partnerCode": "Libancall",
  "loginCustomerId": "57d902ebe85d42329cb51d72",
  "userCode": "001@qq.com",
  "langType": "zh-CN",
  "operateType": "0",
  "currentPage": 1,
  "perPageCount": 3
}

```

The response is a JSON object with the following fields:

```

{
  "streamNo": "LBCALL2016092116475000001",
  "resultCode": "00000000",
  "resultDesc": "成功",
  "data": {
    "perPageCount": 3,
    "currentPage": 1,
    "totalCount": 3,
    "totalPageCount": 3,
    "dataList": [
      {
        "userCode": "Libancall",
        "imei": "",
        "mainType": "ADD",
        "opType": "TRANSFER_AMOUNT",
        "amount": 100,
        "createTime": 1473851927791,
        "goodsNames": null,
        "currencyType": "EUR"
      },
      {
        "userCode": "Libancall",
        "imei": "",
        "mainType": "ADD",
        "opType": "TRANSFER_REWARD",
        "amount": 100,
        "createTime": 1473851939610,
        "goodsNames": null,
        "currencyType": "EUR"
      },
      {
        "userCode": "Libancall",
        "imei": "868740021307048",
        "mainType": "SUBSTRACT",
        "opType": "PROXY_RECHARGE",
        "amount": 5,
        "createTime": 1473851955899,
        "goodsNames": null,
        "currencyType": "EUR"
      }
    ]
  }
}

```

7.3.11 Batch query sub-user list of business customer

7.3.11.1 Call Entity

https://saas.ucloudlink.com/bss/grp/user/QuerySubUserListByBatch?access_token=TGT-266-NHYr7IZutcrGEdJ0qEfbDYWgZzoEgR1MWTMbZlfsQLFuUjIZIG

7.3.11.2 Input Parameter

```

{
  "streamNo": "GLCAM20151113161019000001",
  "partnerCode": "GTBUORG",
  "loginCustomerId": "57ce694cf5b95a5444869213",
  "langType": "en-US",
  "operateType": "1",
  "userCodeList": ["buType@ukeink.com", "147609386043554"]
}

```

7.3.11.3 Output Parameter

```

{

```

```

"streamNo": "GLCAM20151113161019000001",
"resultCode": "00000000",
"resultDesc": "The operation is success.",
"data": [
  {
    "userCode": "147609386043554",
    "customerId": "57fb67a4f5b95a1b2cd18709",
    "imei": "146434231812154",
    "status": "enable",
    "tmiPwd": "",
    "balance": "0.0",
    "currencyType": "EUR",
    "hVersion": "1.0.0",
    "sVersion": "1.0.0",
    "createDate": "1476093860625",
    "remark": null,
    "lockFlag": "1",
    "activationFlag": "0"
  }
]
}

```

7.3.12 Operate offer for sub-user

7.3.12.1 Call Entity

https://saas.ucloudlink.com/bss/grp/goods/SyncOrderRelationFromUP?access_token=TGT-266-NHYr7IZutrGEJ0qEfbDYWgZzoEgR1MWTMbZlfsQLFuUjIZIG

7.3.12.2 Input Parameter

```

{
  "streamNo": "GLCAM20151113161019000001",
  "partnerCode": "GTBUORG",
  "loginCustomerId": "57ce694cf5b95a5444869213",
  "langType": "en-US",
  "operateType": "2",
  "userCode": "buType@ukelink.com",
  "goodsVoList": [
    {
      "orderRelationId": "563b2c3404d0321a149a0ade25b54c82",
      "goodsCode": "HK_500",
      "goodsType": "DISC",
      "effectiveDate": "1472442678402",
      "expiryDate": "1472442686924"
    }
  ]
}

```

7.3.12.3 Output Parameter

```

{
  "streamNo": "GLCAM20151113161019000001",
  "resultCode": "00000000",
  "resultDesc": "The operation is success.",
  "data": null
}

```

7.3.13 Query offer list of sub-user by orderId

7.3.13.1 Call Entity

https://saas.ucloudlink.com/bss/grp/order/QueryOrderRelationByOrderId?access_token=TGT-266-NHYr7IZutcrGEJ0qEfbDYWgZzoEgR1MWTMbZlfsQLFuUjIZIG

7.3.13.2 Input Parameter

```
{
  "streamNo": "GLCAM20180711000000000004",
  "partnerCode": "GTBUORG",
  "loginCustomerId": "57ce694cf5b95a5444869213",
  "langType": "en-US",
  "orderId": "5b62c1c6da811e7a5d445e20"
}
```

7.3.13.3 Output Parameter

```
{
  "streamNo": "GLCAM20180711000000000004",
  "resultCode": "00000000",
  "resultDesc": "Operation successful.",
  "data": [
    {
      "userCode": "8113633xxxx",
      "customerId": "5b4705f7dfd20551f7310xxx",
      "orderRelationId": "5b62c1c6dfd205704e402xxx",
      "goodsId": "5b5af97bda811e7a5d107xxx",
      "goodsName": "Test1",
      "status": "NOT_ACTIVATED"
    },
    {
      "userCode": "8113633xxxx",
      "customerId": "5b4705f7dfd20551f7310xxx",
      "orderRelationId": "5b681765dfd205704e10cxxx",
      "goodsId": "58aa92799f480e1327facxxx",
      "goodsName": "TEST_10M_DP",
      "status": "VALID"
    }
  ]
}
```

7.3.14 Unsubscribe order for sub-user

7.3.14.1 Call Entity

https://saas.ucloudlink.com/bss/grp/order/CancelOrder?access_token=TGT-266-NHYr7IZutcrGEJ0qEfbDYWgZzoEgR1MWTMbZlfsQLFuUjIZIG

7.3.14.2 Input Parameter

```
{
```

```
"streamNo": "GLCAM20180711000000000004",
"partnerCode": "GTBUORG",
"loginCustomerId": "586e0a901ffdba19dd02XXX",
"langType": "en-US",
"orderId": "5b6822c8da811e7a5d165XXX"
}
```

7.3.14.3 Output Parameter

```
{
  "streamNo": "GLCAM20180711000000000004",
  "resultCode": "00000000",
  "resultDesc": "Operation successful.",
  "data": null
}
```

7.3.15 Operate OrderRelation for sub-user

7.3.15.1 Call Entity

https://saas.ucloudlink.com/bss/grp/goods/OperateOrderRelation?access_token=TGT-596285-pAJztraAzAL3VeqhDkKCuabC0t2H63XVV7xJIEJdlj3CicgVI

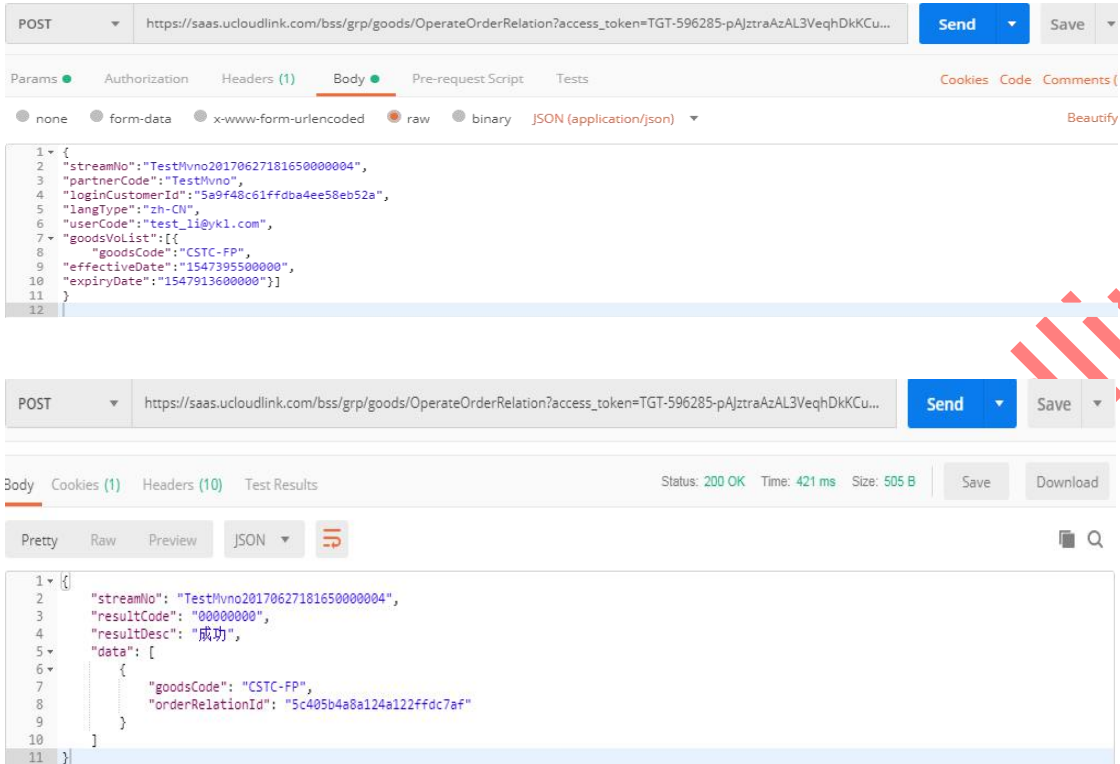
7.3.15.2 Input Parameter

```
{
  "streamNo": "TestMvno2017062718165000000004",
  "partnerCode": "TestMvno",
  "loginCustomerId": "5a9f48c61ffdba4ee58eb52a",
  "langType": "zh-CN",
  "userCode": "test_li@ykl.com",
  "goodsVoList": [
    {
      "goodsCode": "CSTC-FP",
      "effectiveDate": "1547395500000",
      "expiryDate": "1547913600000"
    }
  ]
}
```

7.3.15.3 Output Parameter

```
{
  "streamNo": "TestMvno2017062718165000000004",
  "resultCode": "00000000",
  "resultDesc": "成功",
  "data": [
    {
      "goodsCode": "CSTC-FP",
      "orderRelationId": "5c405b4a8a124a122ffdc7af"
    }
  ]
}
```

7.3.15.4 Test Example



The screenshot shows a REST client interface with a POST request to `https://saas.ucloudlink.com/bss/grp/goods/OperateOrderRelation?access_token=TGT-596285-pAjztraAzAL3VeqhDkKC...`. The request body is a JSON object with the following structure:

```

1 {
2   "streamNo": "TestMvno20170627181650000004",
3   "partnerCode": "TestMvno",
4   "loginCustomerId": "5a9f48c61ffdba4ee58eb52a",
5   "langType": "zh-CN",
6   "userCode": "test_li@ykl.com",
7   "goodsVoList": [
8     {
9       "goodsCode": "CSTC-FP",
10      "effectiveDate": "1547395500000",
11      "expiryDate": "1547913600000"}
12   ]
13 }

```

The response status is 200 OK, with a time of 421 ms and a size of 505 B. The response body is a JSON object with the following structure:

```

1 {
2   "streamNo": "TestMvno20170627181650000004",
3   "resultCode": "00000000",
4   "resultDesc": "成功",
5   "data": [
6     {
7       "goodsCode": "CSTC-FP",
8       "orderRelationId": "5c405b4a8a124a122ffdc7af"
9     }
10   ]
11 }

```

7.3.16 Disable OrderRelation for sub-user

7.3.16.1 Call Entity

`https://saas.ucloudlink.com/bss/grp/goods/DisableOrderRelation?access_token=TGT-596285-pAjztraAzAL3VeqhDkKC...`

7.3.16.2 Input Parameter

```

{
  "streamNo": "TestMvno20170627181650000005",
  "partnerCode": "TestMvno",
  "loginCustomerId": "5a9f48c61ffdba4ee58eb52a",
  "langType": "zh-CN",
  "userCode": "test_li@ykl.com",
  "goodsVoList": [
    {
      "orderRelationId": "5c405b4a8a124a122ffdc7af"
    }
  ]
}

```

7.3.16.3 Output Parameter

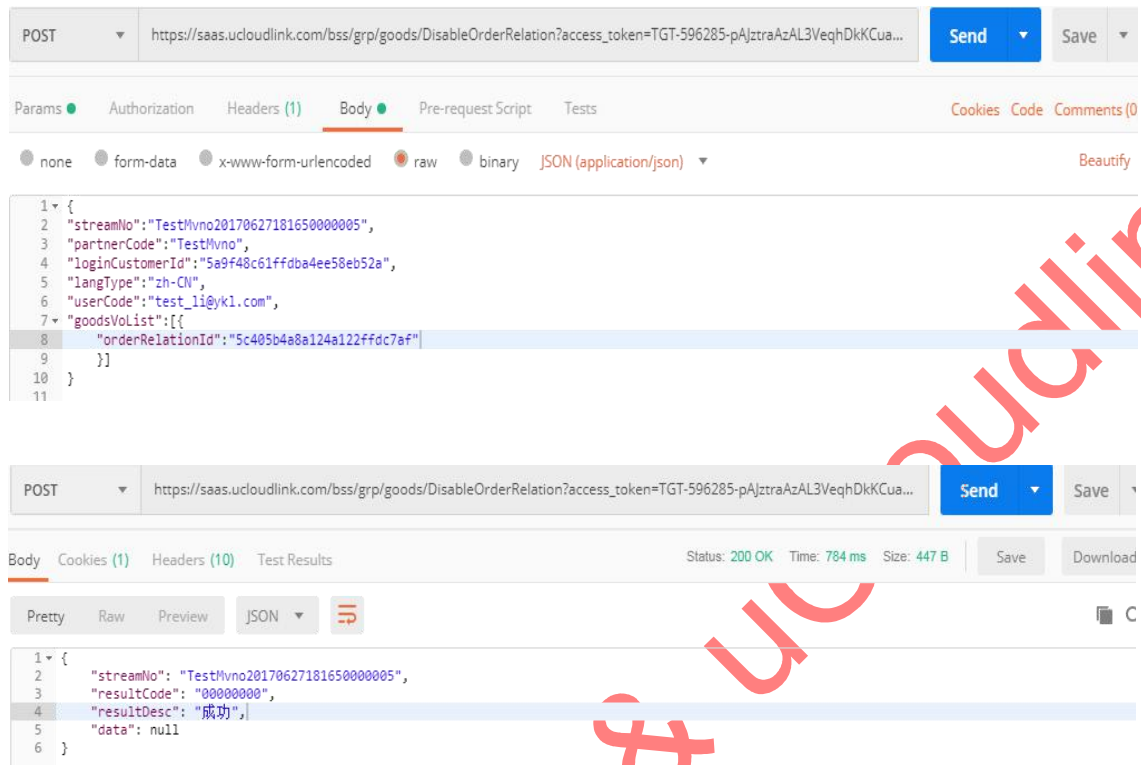
```

{
  "streamNo": "TestMvno20170627181650000005",
  "resultCode": "00000000",
  "resultDesc": "成功",
  "data": null
}

```

}

7.3.16.4 Test Example



The first screenshot shows a POST request to the endpoint `https://saas.ucloudlink.com/bss/grp/goods/DisableOrderRelation?access_token=TGT-596285-pAjztraAzAL3VeqhDkKCua...`. The request body is a JSON object:

```

1 {
2   "streamNo": "TestMvno20170627181650000005",
3   "partnerCode": "TestMvno",
4   "loginCustomerId": "5a9f48c61ffdba4ee58eb52a",
5   "langType": "zh-CN",
6   "userCode": "test_li@ykl.com",
7   "goodsVoList": [{
8     "orderRelationId": "5c405b4a8a124a122ffdc7af"
9   }]
10 }
11

```

The second screenshot shows the response with status `200 OK`, time `784 ms`, and size `447 B`. The response body is a JSON object:

```

1 {
2   "streamNo": "TestMvno20170627181650000005",
3   "resultCode": "00000000",
4   "resultDesc": "成功",
5   "data": null
6 }

```

7.3.17 Update OrderRelation for sub-user

7.3.17.1 Call Entity

`https://saas.ucloudlink.com/bss/grp/goods/UpdateOrderRelation?access_token=TGT-1513871-tCsx3lKrowEZxTwYb9RyOHlCHPYuhQMZ2yExM6fQV9mO4Pq1uU`

7.3.17.2 Input Parameter

```

{
  "streamNo": "TestMvno20190627181650000001",
  "partnerCode": "TestMvno",
  "loginCustomerId": "5a9f48c61ffdba4ee58eb52a",
  "langType": "zh-CN",
  "userCode": "test_li@ykl.com",
  "goodsVoList": [{
    "orderRelationId": "5a9fa5dbe643cb0cedbbea10",
    "effectiveDate": "1567872000000",
    "expiryDate": "1568736000000"}]
}

```

7.3.17.3 Output Parameter

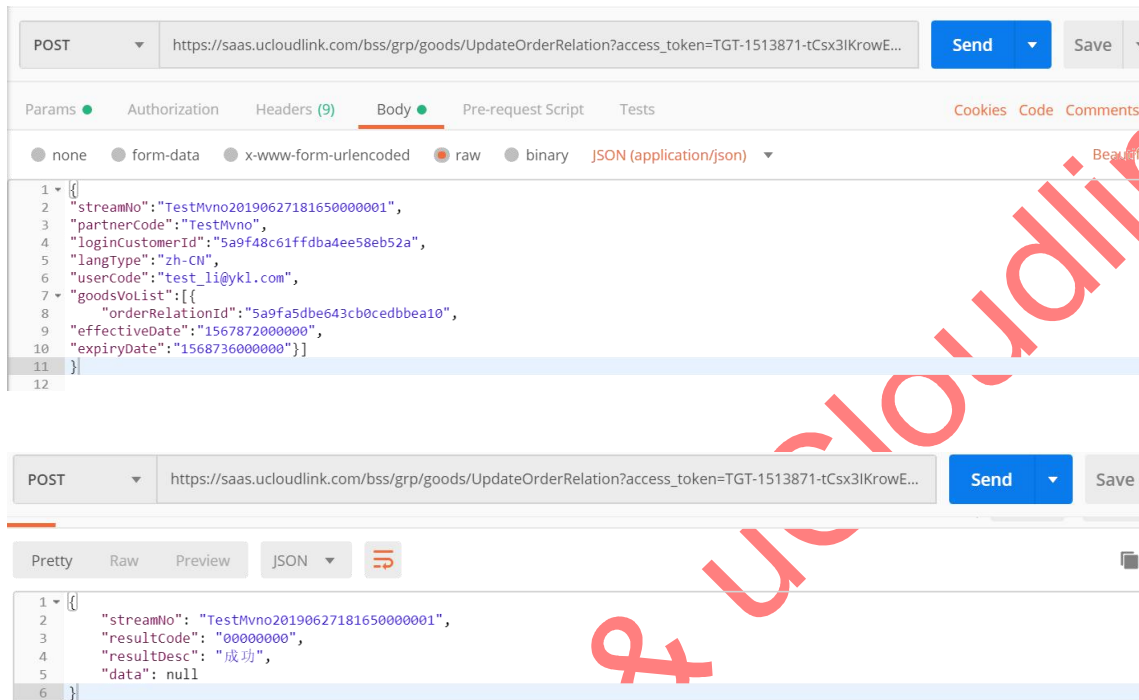
```

{
  "streamNo": "TestMvno20190627181650000001",
  "resultCode": "00000000",

```

```
"resultDesc": "成功",
"data": null
}
```

7.3.17.4 Test Example



The image displays two screenshots of a REST client interface, likely Swagger Client or Postman, showing a POST request and its response.

Top Screenshot (Request):

- Method:** POST
- URL:** `https://saas.ucloudlink.com/bss/grp/goods/UpdateOrderRelation?access_token=TGT-1513871-tCsx3IKrowE...`
- Body Type:** JSON (application/json)
- Body Content:**

```
{
  "streamNo": "TestMvno20190627181650000001",
  "partnerCode": "TestMvno",
  "loginCustomerId": "5a9f48c61ffdba4ee58eb52a",
  "langType": "zh-CN",
  "userCode": "test_li@yk1.com",
  "goodsVolList": [
    {
      "orderRelationId": "5a9fa5dbe643cb0cedbba10",
      "effectiveDate": "1567872000000",
      "expiryDate": "1568736000000"
    }
  ]
}
```

Bottom Screenshot (Response):

- Method:** POST
- URL:** `https://saas.ucloudlink.com/bss/grp/goods/UpdateOrderRelation?access_token=TGT-1513871-tCsx3IKrowE...`
- Response Type:** JSON
- Response Content:**

```
{
  "streamNo": "TestMvno20190627181650000001",
  "resultCode": "00000000",
  "resultDesc": "成功",
  "data": null
}
```